



# Effect of sustained load and seawater and sea sand concrete environment on durability of basalt- and glass-fibre reinforced polymer (B/GFRP) bars

Zike Wang<sup>a,b</sup>, Xiao-Ling Zhao<sup>b,\*</sup>, Guijun Xian<sup>c</sup>, Gang Wu<sup>d</sup>, R.K. Singh Raman<sup>e,f</sup>, Saad Al-Saadi<sup>e</sup>

<sup>a</sup> School of Mechanics and Engineering Science, Zhengzhou University, Zhengzhou 450001, China

<sup>b</sup> Department of Civil Engineering, Monash University, Clayton, VIC 3800, Australia

<sup>c</sup> School of Civil Engineering, Harbin Institute of Technology, Harbin 150090, China

<sup>d</sup> International Institute for Urban Systems Engineering, Southeast University, Nanjing 210096, China

<sup>e</sup> Department of Mechanical and Aerospace Engineering, Monash University, Clayton, VIC 3800, Australia

<sup>f</sup> Department of Chemical Engineering, Monash University, Clayton, VIC 3800, Australia

## ARTICLE INFO

### Keywords:

Seawater and sea sand concrete (SWSSC)

BFRP

GFRP

Sustained stress

Tensile properties

Long-term performance

## ABSTRACT

Recently, in responding to the resource shortage and steel corrosion, the combination of seawater and sea sand concrete (SWSSC) and fibre reinforced polymer (FRP) bars was proposed to replace the traditional reinforced concrete structures in civil engineering. For FRP bars in the real service condition of concrete structure, the synergistic effect of sustained load and corrosion environment needs to be considered. This paper presents an investigation on the durability of basalt- and glass-fibre reinforced polymer bars exposed to a SWSSC environment under different sustained stress levels. Scanning electron microscopy was employed to study the degradation mechanism of FRP bars.

## 1. Introduction

In the year of 2016, the world concrete production was estimated to be as high as 25 billion tonnes [1]. The production of traditional concrete requires use of huge amounts of fresh water and river sands, probably causing their potential shortage. To reduce this resource crisis, seawater and sea sand concrete (SWSSC) could be a good option for civil engineering through replacing fresh water, river sand respectively by seawater and sea sand, especially in isolated islands and coastal areas [2]. It is noted that SWSSC has been found to possess the similar mechanical properties to the traditional concrete [3].

However, the higher chloride content of SWSSC than traditional concrete greatly accelerates the corrosion process of steel reinforcement in concrete [4]. To solve this problem, fibre reinforced polymer (FRP) composites were proposed to be an attractive alternative reinforcement material for SWSSC due to their excellent corrosion resistance [5,6] and other unique properties (e.g., high stiffness/strength-weight ratio, non-magnetic properties, and ease in fabrication) [7–9]. The main application forms of FRP with SWSSC include the internal reinforcements (e.g., FRP reinforcing bars in concrete) [10,11] and the external confinements (e.g., concrete-filled FRP tubes) [12,13]. The traditional FRP bars include glass FRP (GFRP), carbon FRP (CFRP) and aramid FRP (AFRP) [14]. However, GFRP bars are the most commonly used because of their low cost [15], while the use of CFRP bars is still limited due to

its exorbitant cost. More recently, a new and more promising type of FRPs, basalt FRP (BFRP) bars [16,17] were developed, for which the fibres were produced from basalt rocks through a melting process at 1450–1500 °C [18]. BFRP bars generally have superior mechanical properties and chemical stability than E-glass FRP bars [18,19], as well as they are much cheaper than CFRP bars. BFRP bars are also environmentally friendly, non-hazardous and less expensive [18,20,21]. However, for FRP bars in the real service condition of concrete structure, the synergistic effect of sustained load and corrosion needs to be considered [17]. This paper focuses on the durability of BFRP and GFRP bars under sustained load in SWSSC environment.

Since 1990s [15,22–37], the durability of GFRP bars in traditional concrete environment has been extensively studied. The important environmental factors influencing the degradation rate of GFRP bars include pH [15,27], exposure temperature [30], moisture content of concrete [29,33], sustained stress level [23,36]. The degradation rate of GFRP bars also varied with their production parameters, such as bar diameter [25,38], type of fibre [23], type of resin [32,39] and cross-section shape [40]. In concrete environment, the degradation of GFRP bars due to chemical attack by hydroxyl ions (OH<sup>−</sup>) and moisture includes deterioration of matrix, glass fibre and fibre-matrix interface [29,33], and the interface is most prone to such attacks [15]. The sustained stress affects the degradation mechanism of GFRP bars [35,41]. At low stress levels, the degradation of GFRP bars is caused

\* Corresponding author.

E-mail address: [ZXL@monash.edu](mailto:ZXL@monash.edu) (X.-L. Zhao).

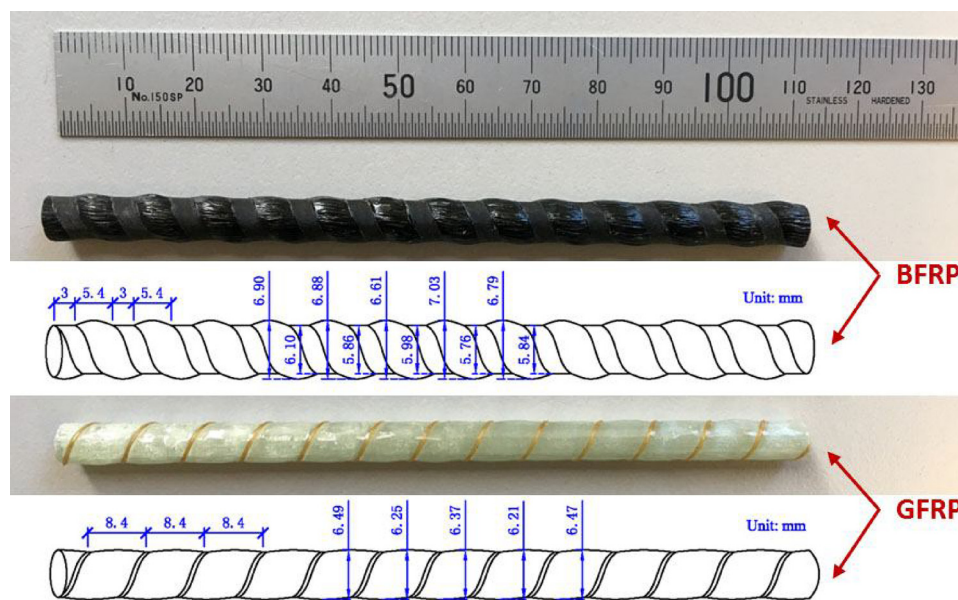


Fig. 1. FRP bars.

only by diffusion of surrounding corrosive medium. At moderate stress levels, the matrix suffers microcracking that facilitates easier access of the corrosive medium to the fibres, thus, accelerating the degradation of the GFRP bars. At high stress levels, the fibres fail by rupture in the shorter periods [35]. More degradation mechanisms are reported in Refs. [30,41–44]. Further, Bemokrane et al. [23] found that the threshold stresses (i.e., the maximum stress below which matrix microcracking does not happen) of GFRP bars in moist concrete were 25%–30% of the ultimate tensile stress. Similar results were also reported by Almusallam and Al-Salloum [26] and Davalos et al. [31]. Moreover, the creep rupture stress limit of GFRP exposed to concrete environment was required as  $0.20 f_{tu}$  ( $f_{tu}$  is the design tensile strength of GFRP) in ACI 440.1R-15 [45]. However, there is little reported on the durability of GFRP bars under sustained load in the SWSSC environment.

Since the interest in BFRPs is relatively recent, the reported durability studies on BFRP bars are rather limited [11,16,17,19,32,46]. These studies showed that the main degradation mechanism of BFRP bars in alkaline environment (e.g., simulated traditional concrete pore solution) is the debonding at the fibre-matrix interface [16,17,32]. The most recent research [11] showed that the degradation of BFRP bars in SWSSC environment may be accelerated due to the chemical reactions involving  $\text{Cl}^-$  and  $\text{Fe}^{2+}$  (Fe-oxide is the content of basalt) [47]. However, there are very limited reported studies on the durability of BFRP bars under sustained load, and they are limited to traditional concrete environment [17,48]. Wu et al. [17] showed that the low stress levels (e.g.,  $\leq 20\%$ ) slightly accelerated degradation of BFRP bars in alkaline solution, while the higher stress levels (e.g., 40% and 60%) hugely accelerated the degradation of BFRP bars.

To the authors' best knowledge, there is still no report on the durability of FRP bars under sustained load in the SWSSC environment. Therefore, to fill this knowledge gap, this study will investigate the synergistic effect of sustained load and SWSSC environment on degradation of the GFRP and BFRP bars. This study is an extension of that on durability of FRP bars under zero-stress condition in seawater and sea sand concrete environment [11]. In this study, a total of 37 tensile specimens of each of BFRP and GFRP bars were subjected to different stress levels (i.e., 20%, 30% and 40% of the average ultimate tensile strength) while exposed to SWSSC environment at 25, 40 and 55 °C, i.e., constitute accelerated corrosion conditions. The scanning electron microscopy (SEM) was used to investigate the degradation mechanism of FRP bars. Finally, the long-term behaviours of BFRP bars under zero-

stress and 20% stress levels in SWSSC environment were predicted using the prediction models based on the Arrhenius theory.

## 2. Experimental program

### 2.1. Materials

As shown in Fig. 1, two kinds of FRP bars (i.e., BFRP and GFRP) with 6 mm diameter were chosen for this study. BFRP bars were the same as the materials in the published literatures [10,11,17], and GFRP bars were the same as those in the authors' previous studies [10,11]. BFRP bars were manufactured by Jiangsu Green Materials Vally New Material T&D Co., Ltd (Nanjing, China). GFRP bars were manufactured by Harbin FRP Institute (Harbin, China). BFRP and GFRP bars were both produced with a pultrusion process using the same epoxy resin with the corresponding fibres fully consistent with the manufacturing parameters provided in the authors' earlier publications [10,11]. The surface of BFRP bars was ribbed with a nylon laminate during pultrusion. The surface of GFRP bars was helically wrapped with a Kevlar fibre bundle with 0.4 mm diameter during pultrusion. These rough treatments caused some unevenness in the diameter of the FRP bars. The average cross-sectional areas of specimens for the immersion testing were 28.52 mm<sup>2</sup> and 31.16 mm<sup>2</sup> for BFRP and GFRP bars, respectively. Accordingly, the nominal diameters of BFRP and GFRP bars used in the calculation of the mechanical properties were 6.0 mm and 6.3 mm, respectively. The production parameters of the FRP bars are listed in Table 1.

### 2.2. Tensile and physical properties

The basic tensile and physical properties of BFRP and GFRP bars supplied by the manufacturers are listed in Table 2. It is clear that for BFRP bars, 20%, 30% and 40% of the average ultimate tensile strength are equivalent to 21%, 32% and 43% of their guaranteed tensile strength, respectively. Similarly, for GFRP bars, 20%, 30% and 40% of the average ultimate tensile strength are equivalent to 22%, 33% and 43% of their guaranteed tensile strength, respectively. In order to have a direct comparison with the results of previous studies [17,24,26,37], the sustained loads applied on FRP bars in this study were based on the average ultimate tensile strength of BFRP and GFRP bars.

$T_g$  of cured resin used for BFRP and GFRP bars was characterized using DMA Q800 (TA Instruments, USA) in a single cantilever mode

**Table 1**  
Production parameters of FRP bars.

| Type | Production diameter (mm) | Reinforced fibre |                            |                        |                       |                                    | Resin matrix     |                     |
|------|--------------------------|------------------|----------------------------|------------------------|-----------------------|------------------------------------|------------------|---------------------|
|      |                          | Brand            | Diameter ( $\mu\text{m}$ ) | Tensile strength (MPa) | Tensile modulus (GPa) | Density ( $\text{g}/\text{cm}^3$ ) | Epoxy            | Hardener            |
| BFRP | 6                        | 1200 Tex         | 13                         | 3800–4840              | 93.1–110              | 2.65                               | Bisphenol-A E-51 | MeHHPA <sup>a</sup> |
| GFRP | 6                        | E-glass 9600 Tex | 28                         | 2200                   | 76                    | 2.54                               | Bisphenol-A E-51 | MeHHPA              |

Note: a: MeHHPA represents Methylhexahydrophthalic Anhydride.

**Table 2**  
Tensile and physical properties of reference BFRP and GFRP bars (as provided by manufacturers).

| Property            |                                                                                   | Symbol                                      | BFRP bars        | GFRP bars      |
|---------------------|-----------------------------------------------------------------------------------|---------------------------------------------|------------------|----------------|
| Tensile properties  | Ultimate tensile strength (MPa)                                                   | $f_{u,ave}$                                 | $1398 \pm 30$    | $1050 \pm 27$  |
|                     | Guaranteed tensile strength (MPa)                                                 | $f_{fu}^* = f_{u,ave} - 3\sigma$            | 1308             | 969            |
|                     | Environmental reduction factor (ACI 440.1R-15)                                    | $C_E$                                       | 0.7 <sup>a</sup> | 0.7            |
|                     | Design tensile strength (ACI 440.1R-15) (MPa)                                     | $f_{fu} = C_E f_{fu}^*$                     | 916              | 678            |
|                     | Modulus of elasticity (GPa)                                                       | $E_{f,ave}$                                 | $46 \pm 0.4$     | $36 \pm 0.9$   |
|                     | Ultimate strain (%)                                                               | $\epsilon_{u,ave}$                          | $3.0 \pm 0.07$   | $3.0 \pm 0.08$ |
|                     | Guaranteed strain (%)                                                             | $\epsilon_u^* = \epsilon_{u,ave} - 3\sigma$ | 2.79             | 2.76           |
|                     | Design strain (%)                                                                 | $\epsilon_u = C_E \epsilon_u^*$             | 1.95             | 1.93           |
| Physical properties | Glass transition temperature ( $^{\circ}\text{C}$ )                               | $T_g$                                       | $\sim 130^b$     | 140            |
|                     | Fibre content by volume (%)                                                       | $V_f$                                       | 65.0             | 62.8           |
|                     | Fibre content by weight (%)                                                       | $W_f$                                       | 80.5             | 78.3           |
|                     | Transverse coefficient of thermal expansion ( $\times 10^{-6}/^{\circ}\text{C}$ ) | $\alpha_T$                                  | 21.7 [32]        | 22.7 [32]      |

Note: a: the  $C_E$  for BFRP bars was assumed to be 0.7 in this study.

b:  $T_g$  of cured matrix used inside BFRP and GFRP bars were measured in this study.

with a support span length of 17.5 mm, heating rate of  $5^{\circ}\text{C}/\text{min}$  from room temperature to  $200^{\circ}\text{C}$ , amplitude of  $15\mu\text{m}$ , and frequency of 1 Hz. The peak of  $\tan \delta$  curve was adopted as the glass transition temperature ( $T_g$ ).  $T_g$  of the cured resin of BFRP bars was determined to be  $\sim 130^{\circ}\text{C}$ , and  $140^{\circ}\text{C}$  for GFRP. The difference in  $T_g$  values for BFRP and GFRP bars are caused by the different curing parameters employed for the two during their manufacturing [10,11]. The fibre contents of BFRP and GFRP bars were characterized using the matrix burning-off method using a muffle furnace, in accordance with ASTM D3171-15 [49]. The fibre volumes of BFRP and GFRP bars were determined to be 65.0% and 62.8%, respectively. For the similar type of FRP bars, Bemokrane et al. [32] assumed the transverse coefficients of thermal expansion for the BFRP and GFRP bars to be  $21.7 \times 10^{-6}/^{\circ}\text{C}$  and  $22.7 \times 10^{-6}/^{\circ}\text{C}$  respectively.

### 2.3. Tensile test method

Tensile tests on BFRP and GFRP bars were conducted in accordance with the modified ACI-440.3R B.2-04 [50] using a 500 kN capacity Baldwin machine. The reason for choosing the earlier code (ACI 440.3R B.2-04 [50]) in this study, instead of the more recent ones (e.g., ACI 440.3R-12 [51] or ASTM D7205/D7205M-06(2016) [52]) was to enable a direct comparison with the tensile results in previous literatures [11,17]. The tensile specimen was cut to a length of 760 mm, and the length of the test section was 260 mm. Each end of an FRP specimen was coated with silica sands and then mounted in 250 mm-long steel tubes (with threads on the outer surface) grouted with an epoxy resin mortar matrix. Aluminium caps were used to close both ends of the steel tubes and to keep the bar along the centreline of the tube during anchor installation. The anchoring specimen was cured at room temperature for at least one week before the tensile test. A tension rate of 2 mm/min was adopted. The longitudinal deformation of FRP specimens during the tensile tests was monitored with a string potentiometer. For BFRP and GFRP bars, five reference specimens were tested (with two conditioned specimens tested for each group). The reason to test only two conditioned specimens in this study is that both specimen results are

rather close to each other. The coefficient of variance of tensile strengths of most groups was less than 10%. This test deviation was acceptable for characterisation of the difference in trend of BFRP and GFRP bars during the exposure. However, the future researchers are encouraged to test at least five conditioned specimens based on the requirement of ACI 440.3R-12 [51] or ASTM D7205/D7205M-06(2016) [52]. More details of the tensile test can be found in Ref. [11].

### 2.4. Simulated seawater and sea sand concrete (SWSSC) solution

Based on the findings of studies on unstressed BFRP and GFRP bars in two kinds of simulated SWSSC pore solutions [10,11,15], a more moderate solution with a lower pH, i.e., a solution simulating high performance seawater and sea sand concrete (HP-SWSSC) pore solution, was adopted in this study to investigate the durability of stressed BFRP and GFRP bars under the combined effect of stress and the SWSSC environment. The simulated HP-SWSSC solution was composed of 0.6 g of NaOH, 1.4 g of KOH, 0.037 g  $\text{Ca}(\text{OH})_2$  and 35 g NaCl in 1 L of distilled water, and the measured pH was 12.7. The detailed composition of HP-SWSSC solution is described in Table 3.

### 2.5. Pre-loading setup and test procedure

In this study, a specific setup was developed, which is a modification of the test setup in Wu et al. [17] to apply the sustained load on BFRP and GFRP bars during their exposure to HP-SWSSC solution. Three stress levels (i.e., 20%, 30% and 40% of the average ultimate

**Table 3**  
Compositions of simulated HP-SWSSC pore solution.

| Type              | Quantities (gram per litre) |     |                          |      |      |
|-------------------|-----------------------------|-----|--------------------------|------|------|
|                   | NaOH                        | KOH | $\text{Ca}(\text{OH})_2$ | NaCl | pH   |
| HP-SWSSC solution | 0.6                         | 1.4 | 0.037                    | 35   | 12.7 |



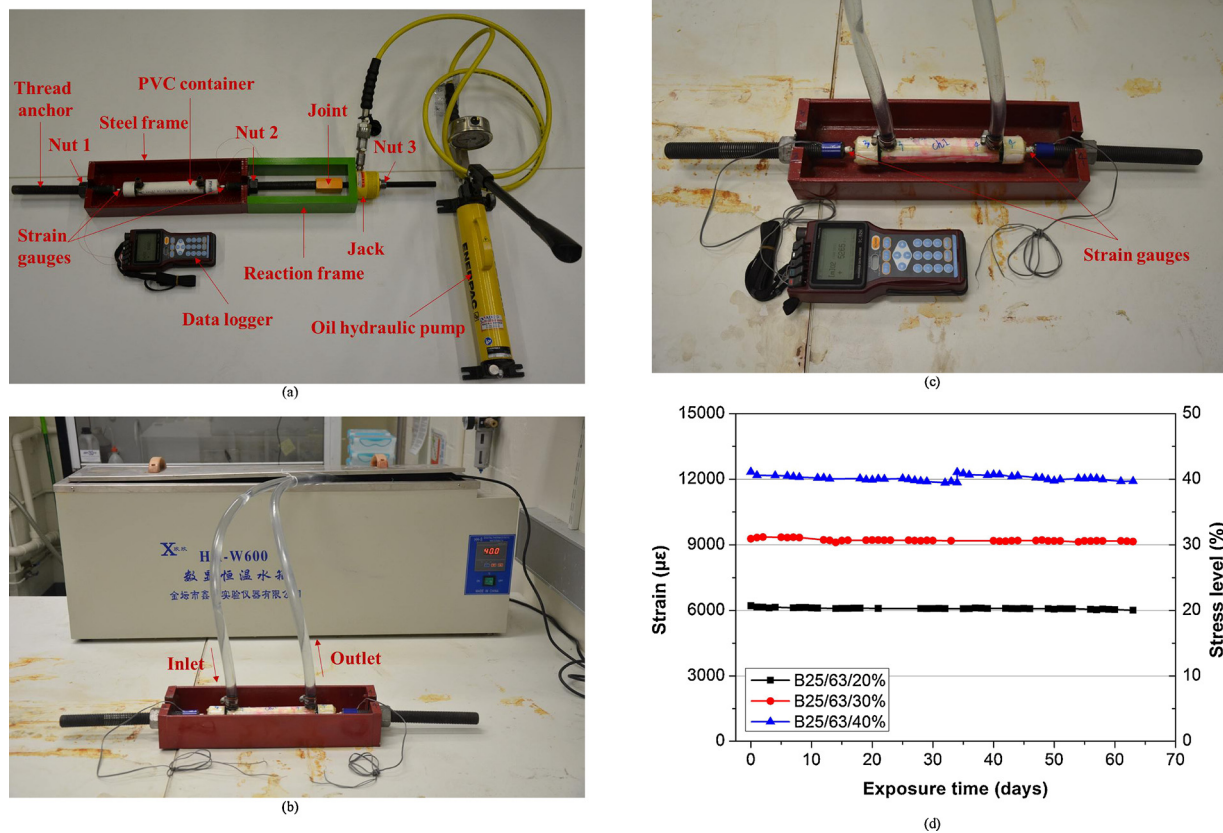


Fig. 2. Preparation of conditioned FRP tensile specimens: (a) pre-loading set-up; (b) HP-SWSSC solution exposure system; (c) data logger for strain measurement; and (d) typical collected strain change with exposure time.

tensile strength of BFRP and GFRP bars) were applied.

The whole test procedure for conditioned FRP specimens before the tensile test mainly includes:

**Stage I (pre-loading of FRP):** Each FRP specimen was pre-loaded using the devices shown in Fig. 2(a) according to the following steps. Step (1): For loading of FRP, the middle part of FRP bar was inserted inside a 30 mm-diameter and 240 mm-long corrosion cell (PVC container), and then both ends of FRP bar were anchored inside the steel tubes with an epoxy resin mortar matrix. Two strain gauges were then glued on the intact area of FRP bar (i.e., the gap between PVC container and thread anchors). Considering the variation in Young's modulus for different FRP specimens, the stress-strain relationship of each FRP tensile specimen was calibrated using the Baldwin machine and data logger before the pre-loading of FRP bars. Step (2): The tensile specimen from Step (1) was placed in the "U" shape notches of steel frame, and assembled with a reaction frame and hydraulic jack by some auxiliary components (i.e., Nuts 1, 2, 3 and joint), and two strain gauges were in turn connected with the data logger. In this step, since the FRP specimen was in the zero-stress state, the measured values of strain gauges should be around zero. Step (3): Nuts 1 and 3 were both tightened with a wrench and then the axial load on FRP specimen was applied by the hydraulic jack with the aid of a reaction frame and joint. In this step, the measured strain increases with the increasing load. Step (4): After the required stress level, as calculated when the measured strain value was reached, Nut 2 was tightened and the hydraulic jack was unloaded. Finally, all the other auxiliary devices were removed.

**Stage II (FRP exposure to solution):** After the FRP specimen was locked with the required stress level, both ends of the PVC container were tightly sealed using the epoxy adhesive. For exposure of FRP specimen to the given corrosive condition, the HP-SWSSC solution was circulated through the PVC container at constant temperature that was maintained using a thermostat bath, as shown in Fig. 2(b). More details

of the corrosive solution exposure system can be found in a previous study [11].

**Stage III (consistent stress control of FRP):** During the entire test duration, the strains on the intact area of FRP specimen (i.e., area outside the container) were measured periodically with the data logger, as shown in Fig. 2(c). As the intact area of FRP bars was not exposed to HP-SWSSC solution throughout, the mechanical behaviour (e.g., Young's modulus, stress-strain relationship) in the intact area remained unchanged during the entire test duration. Therefore, the consistent strain level corresponds to the consistent stress level for the FRP bars. When the measured strain dropped below the required strain/stress level, the load was increased to the required level, following Steps (2)–(4) in Stage I. The typical change in strain/stress level with the exposure time for conditioned BFRP bars is shown in Fig. 2(d).

After the simultaneous loading and corrosion exposure for the defined duration, the pre-load on FRP specimen was released and the PVC container was removed and the specimen was subjected to the tensile test according to the modified 440.3R B.2-04 [50].

## 2.6. Definition of specimen label

In this study, a total of 37 tensile specimens of each of BFRP and GFRP bars were tested at three exposure temperatures (25 °C that refers to room temperature, 40 °C and 55 °C) for three exposure periods (21, 42 and 63 days). The specimen label system was also defined as follows:

For unconditioned specimens, "BR" represents the BFRP reference specimen and "GR" represents the GFRP reference specimen.

For a typical conditioned specimen, "B25/21/20%" represents the BFRP specimen under 20% stress level after pre-exposure to HP-SWSSC solution at 25 °C for 21 days. In the label "B25/21/20%", "B" refers to the BFRP bar; the first figure "25" means the exposure temperature of 25 °C; the second figure "21" means the exposure time of 21 days and

**Table 4**

Experimental tensile properties of reference and conditioned BFRP specimens exposed to HP-SWSSC solution.

| Specimen label | Tensile strength |               |                     | Elastic modulus |        |
|----------------|------------------|---------------|---------------------|-----------------|--------|
|                | Mean (MPa)       | Retention (%) | CV <sup>a</sup> (%) | Mean (GPa)      | CV (%) |
| BR             | 1357             | 100           | 4.2                 | 42              | 5.2    |
| B25/21/20%     | 1255             | 92.4          | 5.9                 | 41              | 3.7    |
| B25/42/20%     | 1276             | 94.0          | 4.2                 | 44              | 2.3    |
| B25/63/20%     | 1261             | 92.9          | 7.9                 | 44              | 4.8    |
| B40/21/20%     | 1200             | 88.4          | 3.2                 | 41              | –      |
| B40/42/20%     | 1219             | 89.8          | 0.8                 | 46              | 4.0    |
| B40/63/20%     | 1109             | 81.7          | 4.0                 | 43              | 6.1    |
| B55/21/20%     | 945              | 69.6          | 10.4                | 46              | 1.9    |
| B55/42/20%     | 811              | 59.8          | 8.7                 | 47              | 0.9    |
| B55/63/20%     | 748              | 55.1          | 7.2                 | 44              | 7.9    |
| B25/21/30%     | 1129             | 83.1          | 2.2                 | 45              | 6.7    |
| B25/42/30%     | 1024             | 75.4          | 7.1                 | 45              | 2.6    |
| B25/63/30%     | 1071             | 78.9          | 4.6                 | 46              | 3.7    |
| B25/21/40%     | 804              | 59.2          | 3.0                 | 38              | 2.4    |
| B25/42/40%     | 733              | 54.0          | 6.0                 | 40              | 4.4    |
| B25/63/40%     | 586              | 43.2          | 1.4                 | –               | –      |
| B40/21/40%     | Rupture          | –             | –                   | –               | –      |

Note: a: Coefficient of variation.

“–” represents “not available”.

The tensile data of BFRP bars under zero-stress condition refer to [11].

the last figure “20%” means the stress level of 20%. Similarly, “G25/21/20%” represents the GFRP specimen under 20% stress level after pre-exposure in HP-SWSSC solution at 25 °C for 21 days.

For the referred data from the previous research on BFRP and GFRP bars under zero-stress condition [11], “B32/21” represents the BFRP specimen after pre-exposure to HP-SWSSC solution at 32 °C for 21 days. In the label “B32/21”, “B” refers to the BFRP bar; the first figure “32” means the exposure temperature of 32 °C and the second figure “21” means the exposure time of 21 days. Similarly, “G32/21” represents the GFRP specimen after pre-exposure to HP-SWSSC solution at 32 °C of 21 days.

Similarly, for the referred data for BFRP bars in alkaline solution [17], “AB40” represents the BFRP specimens in zero-stress condition after pre-exposure to alkaline solution at 40 °C. In the label “AB40”, “A” means the alkaline solution used in the literature [17]. “AB40/20%” represents the BFRP specimens under 20% stress level after pre-exposure to alkaline solution at 40 °C.

## 2.7. Scanning electron microscopy (SEM)

The morphologies of the cross section and longitudinal section for reference and conditioned BFRP and GFRP specimens were both characterized using a JEOL 7001 F field emission scanning electron microscope (SEM, Japan). All the SEM specimens were chosen before the tensile test of FRP bars.

## 3. Results and discussion on tensile properties

All the tensile results of reference and conditioned BFRP and GFRP bars are summarized in Tables 4 and 5 respectively. It is noted that the tested tensile strength and elastic modulus (i.e.,  $1357 \pm 57$  MPa and  $42 \pm 2.2$  GPa) of reference BFRP are both respectively higher than those of GFRP bars (i.e.,  $1054 \pm 33$  MPa and  $36 \pm 0.8$  GPa). This is mainly attributed to the considerably superior mechanical properties of basalt fibre than those of glass fibre, as listed in Table 1.

### 3.1. Tensile strength retention

#### 3.1.1. BFRP bars

The effect of stress level during exposure on the subsequent tensile

**Table 5**

Experimental tensile properties of reference and conditioned GFRP specimens exposed to HP-SWSSC solution.

| Specimen label | Tensile strength |               |                     | Elastic modulus |        |
|----------------|------------------|---------------|---------------------|-----------------|--------|
|                | Mean (MPa)       | Retention (%) | CV <sup>a</sup> (%) | Mean (GPa)      | CV (%) |
| GR             | 1054             | 100           | 3.1                 | 36              | 2.6    |
| G25/21/20%     | 984              | 93.4          | 1.0                 | 35              | 3.9    |
| G25/42/20%     | 988              | 93.7          | 2.2                 | 35              | 1.0    |
| G25/63/20%     | 956              | 90.7          | 4.6                 | 37              | 2.3    |
| G40/21/20%     | 915              | 86.8          | 6.6                 | 36              | 4.3    |
| G40/42/20%     | 865              | 82.0          | 9.8                 | 36              | 2.0    |
| G40/63/20%     | 781              | 74.1          | 12.5                | 35              | 0.4    |
| G55/21/20%     | 724              | 68.7          | 8.5                 | 38              | 0.7    |
| G55/42/20%     | 773              | 73.4          | 7.1                 | 36              | 0.6    |
| G55/63/20%     | 617              | 58.6          | 9.9                 | 35              | 2.8    |
| G25/21/30%     | 931              | 88.3          | 0.8                 | 35              | 1.6    |
| G25/42/30%     | 889              | 84.3          | 5.6                 | 35              | 1.5    |
| G25/63/30%     | 894              | 84.8          | 3.5                 | 35              | 1.2    |
| G25/21/40%     | 832              | 78.9          | 11.0                | 36              | 2.5    |
| G25/42/40%     | 834              | 79.2          | 7.0                 | 34              | 1.0    |
| G25/63/40%     | 717              | 68.0          | 8.4                 | 32              | 7.2    |
| G40/21/40%     | Rupture          | –             | –                   | –               | –      |

Note: a: Coefficient of variation.

“–” represents “not available”.

The tensile data of GFRP bars under zero-stress condition refer to [11].

strength retention of conditioned BFRP bars is shown in Fig. 3. As shown in Fig. 3(a), for a given exposure temperature and time, it is clear that with the increasing stress level during exposure, the tensile strength retention of BFRP bars obviously reduces. After exposure to HP-SWSSC solution at room temperature for 63 days, the tensile strength retention of BFRP bars under zero-stress, 20%, 30% and 40%

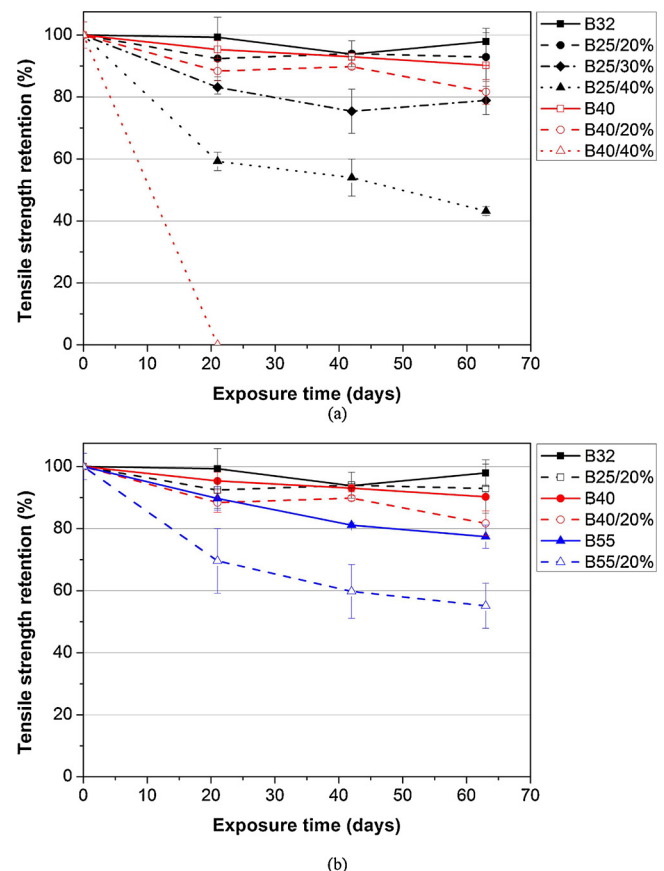


Fig. 3. Effect of conditioning in HP-SWSSC solution on the tensile strength retention of BFRP bars: (a) different stress levels and (b) 20% stress level.

stress levels are 97.9%, 92.9%, 78.9% and 43.2% respectively. The elevated temperature exposure causes more degradation of BFRP. After exposure to HP-SWSSC solution at 40 °C at 20% stress level for 63 days, the tensile strength retention of BFRP specimen is 81.7%. But, at the 40% stress level, the BFRP specimen ruptured rapidly in less than 21 days while exposed to HP-SWSSC solution at 40 °C. This suggests the 40% stress level to be too high for BFRP bars in HP-SWSSC environment. In addition, Fig. 3(b) shows that the 20% stress level can cause more degradation to BFRP bars when exposed to HP-SWSSC solution than the unstressed BFRP bars at room temperature, 40 °C or 55 °C, and the degradation increases with the increase in the exposure temperature. This shows that the combined effect of temperature and stress obviously accelerates the degradation of BFRP bars in HP-SWSSC solution.

In addition, it is noted that the tensile strength retention in some cases of BFRP (i.e., B25/20%, B25/30% and B40/20%) decreases significantly after 21 days of exposure, but lightly increases after 42 or 63 days of exposure. This phenomenon of “strength resurrection” for BFRP bars has been attributed to post-curing of matrix caused by the moisture ingress, in an earlier study [53]. Similar findings are also reported for pultruded FRP bars/plates immersed in seawater and alkaline solution [32,53].

### 3.1.2. GFRP bars

The effect of stress level on tensile strength retention of conditioned GFRP bars is shown in Fig. 4. As shown in Fig. 4(a), similar to the results of BFRP bars, the higher the stress level, the lesser is the tensile strength retention of GFRP bars. After exposed to HP-SWSSC solution at room temperature for 63 days, the tensile strength retention of GFRP bars under zero-stress, 20%, 30% and 40% stress level are 97.9%, 90.7%, 84.8% and 68.0% respectively. Again, at the 40% stress level,

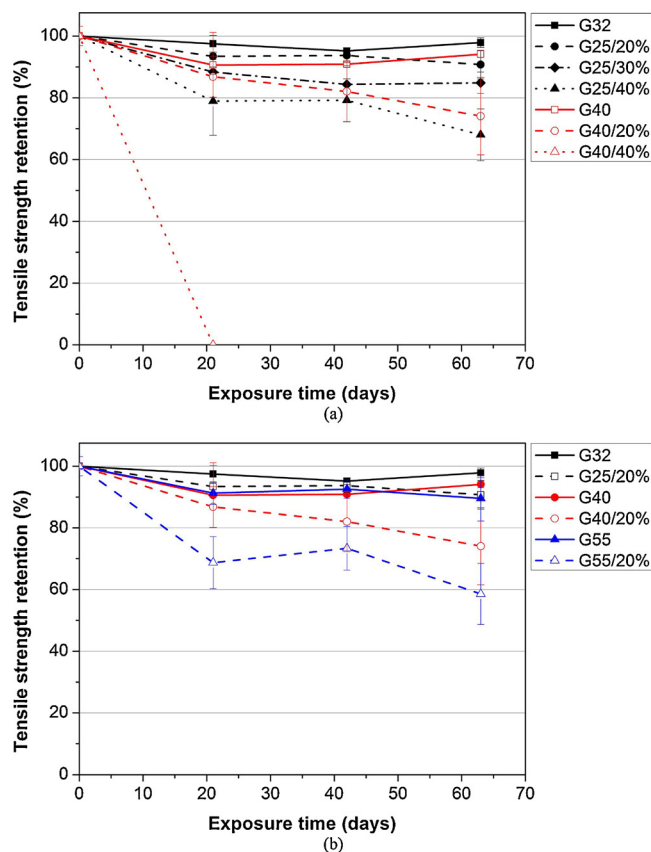


Fig. 4. Effect of conditioning in HP-SWSSC solution on the tensile strength retention of GFRP bars: (a) different stress levels and (b) 20% stress level.

the GFRP specimen ruptured prematurely in less than 21 days of exposure to HP-SWSSC solution at 40 °C, again, suggesting that the 40% stress level is too high for GFRP bars in HP-SWSSC environment. Similarly, Fig. 4(b) shows that 20% stress level can obviously cause more degradation of GFRP bars than unstressed GFRP bars at each of the three exposure temperatures. The synergistic effect of temperature and stress in accelerating the degradation rate also exists for GFRP bars. The phenomenon of “strength resurrection” is also found for some cases of GFRP (i.e., G25/20%, G25/30% and G55/20%), which is again attributed to the post-curing of matrix.

### 3.1.3. Comparison of BFRP and GFRP

From the comparison of Figs. 3 and 4 (and data in Tables 3 and 4), under the condition of 20% stress level and pre-exposure to HP-SWSSC at 25 °C, the degradation in tensile strength retention of BFRP bars is similar to that of GFRP bars. However, with the increase in exposure temperature and/or stress level, the BFRP bars degrade faster than the GFRP bars (under the identical condition of HP-SWSSC exposure, stress level, exposure temperature and time), with the exception of the condition of 20% stress at 40 °C. A comparison of surface appearance confirms the GFRP bars to have suffered less degradation than BFRP bars (as described in Section 4.1). Based on above results, the durability of stressed GFRP bars seems to be superior to that of stressed BFRP bars during exposure to the same HP-SWSSC condition. This is consistent with the reported findings from the authors’ group, on the unstressed GFRP and BFRP bars exposed to HP-SWSSC solution [10,11]. The typical moisture absorption/desorption test results shown in Table 6 [10] suggest greater moisture uptake by BFRP than GFRP. It is crucial to note that the reported chemical reaction between  $\text{Cl}^-$  and  $\text{Fe}^{2+}$  that is exclusive to basalt fibre because of Fe-oxide content of basalt [11,47]. This reaction facilitates greater degradation BFRP bars than GFRP bars in HP-SWSSC solution.

As shown in Figs. 3 and 4, the degradation of both BFRP and GFRP bars was further enhanced after more than 63 days of exposure HP-SWSSC solution. This is because the interlaminar shear strength of BFRP and GFRP bars immersed in HP-SWSSC solution has been shown to continually decrease after exposure to corrosive medium for > 63 days, in another study from the authors’ group [10]. The continued increase in degradation of BFRP and GFRP bars with the increasing exposure time also conforms the basic assumption of long-term prediction models of FRP bars adopted in the present study (as described in Section 5.1).

### 3.1.4. Comparison of present and literature data

Fig. 5 presents the comparison of experimental data for BFRP and

Table 6

Moisture absorption and desorption test results of unstressed BFRP and GFRP bars in HP-SWSSC solution (as extracted from Ref. [10]).

| Specimen label | Weight gain/change percent (%) |                                    | Real moisture absorption percent (%) |
|----------------|--------------------------------|------------------------------------|--------------------------------------|
|                | After exposure for 84 days     | After drying at 60 °C <sup>a</sup> |                                      |
| B25            | 0.10 (0.01)                    | −0.08 (0.02)                       | 0.18 <sup>b</sup>                    |
| B40            | 0.46 (0.06)                    | −0.28 (0.04)                       | 0.74                                 |
| B55            | 0.97 (0.28)                    | −1.73 (0.39)                       | 2.70                                 |
| G25            | 0.17 (0.03)                    | −0.05 (0.03)                       | 0.22                                 |
| G40            | 0.11 (0.02)                    | −0.16 (0.05)                       | 0.27                                 |
| G55            | 0.03 (0.09)                    | −0.56 (0.14)                       | 0.59                                 |

Note: a: the desorption test was carried out for conditioned specimens after exposed to HP-SWSSC solution for 84 days.

b: the value of real moisture absorption percent was calculated based on the results of absorption and desorption tests, i.e.,  $0.18 = 0.10 + 0.08$ .

The values in the parentheses represent the standard deviation.

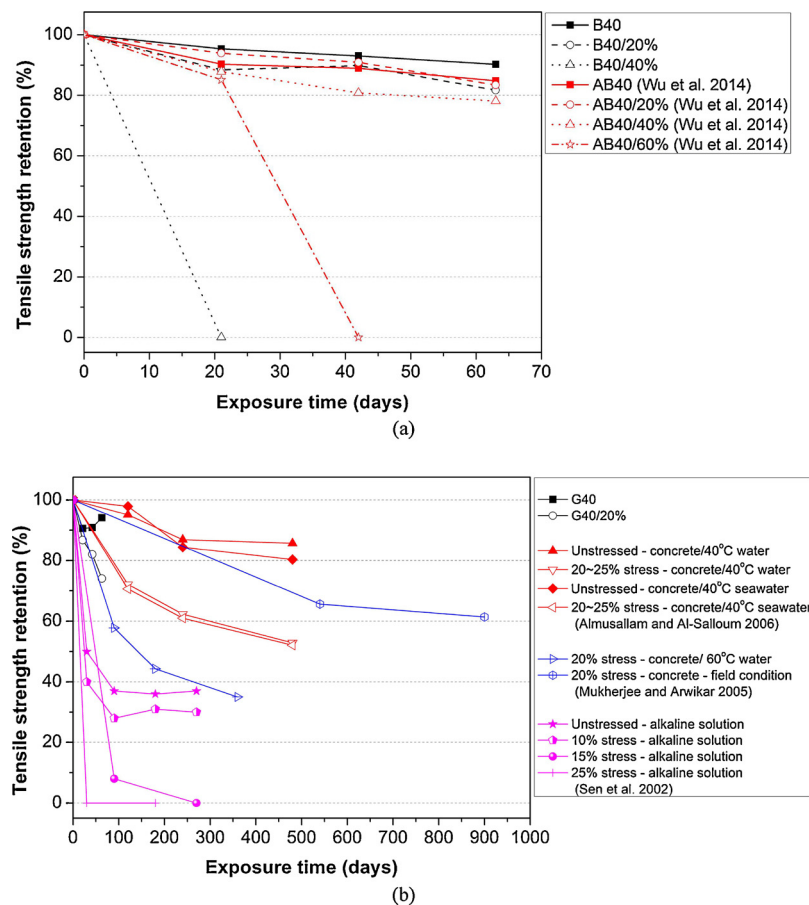


Fig. 5. Comparison of experimental data between this study and previous literatures: (a) BFRP bars and (b) GFRP bars.

GFRP bars from this study with those reported in the literature. To date, only one paper seems to have focused on the tensile strength retention of BFRP bars at different stress levels in an alkaline solution [17]. In this study the BFRP bars identical to those in Wu et al.'s [17] study were used. From the comparison shown in Fig. 5(a), it is clear that the 20% stress level causes a relatively lower degree of degradation to BFRP bars than the higher stress levels (40% or 60%) in HP-SWSSC as well as alkaline solutions. In addition, Fig. 5(a) shows a strange phenomenon, i.e., under the same stress level (e.g., 20% or 40%), the degradation of BFRP bars in HP-SWSSC solution is much quicker than that in alkaline solution even though the alkaline solution (pH = 12.9–13.1) is more aggressive than HP-SWSSC solution (pH = 12.7). The only way this could be explained is the likely loss of stress level during the exposure of BFRP bars in alkaline solution – it is noted that the strain or stress level of BFRP bars was not continuously monitored in the reported study [17]. This explanation is readily supported by the fact that in the present study it became necessary to compensate for drop in the stress level during exposure, by tightening Nut 2 (as described in Section 2.5).

Fig. 5(b) compares the tensile stress retention data from this study and those in the literature, for GFRP bars in various corrosive environments. Sen et al. [24] investigated E-glass/vinylester GFRP bars at three stress levels (10%, 15% and 25%) exposed to high alkalinity (pH = 13.5) at room temperature for 9 months. Mukherjee and Arwikaar [37] investigated E-glass/vinylester GFRP bars in two kinds of corrosive environments. In the first, GFRP bars were embedded in a concrete beam at 20% stress level and then conditioned in 60 °C water for 12 months. For the second environment, GFRP bars were embedded in concrete beam at 20% stress level and then conditioned in natural weathering outdoors (i.e., field condition) in the industrial city of

Mumbai (Bombay, India) for 30 months. Almusallam and Al-Salloum [26] investigated E-glass/vinylester GFRP bars embedded in concrete beam at 20–25% stress level and then conditioned in 40 °C water (or seawater) for 16 months.

As shown in Fig. 5(b), the 20% stress level can effectively cause more degradation to stressed GFRP bars than to the unstressed GFRP bars in various corrosive environments. This is consistent with the fact that the recommended creep rupture stress limit of GFRP bars generally is  $0.20 f_{fu}$  ( $f_{fu}$  is the design tensile strength of FRP) in ACI 440.1R-15 [45]. Fig. 5(b) also shows that the degradation of GFRP bars embedded in concrete is much slower than that of GFRP bars completely exposed to simulated concrete pore solutions. This conclusion is consistent with other studies [29,33,54]. This can be explained by the difference in corrosion mechanisms [54]: for GFRP embedded in a moisture-saturated concrete, the erosion of GFRP bars by inner  $H_2O/OH^-$  can only be a dense-point type, and the flow of  $H_2O/OH^-$  is hindered by the aggregates. Meanwhile, the penetration of ions further into the interior of GFRP bars is further hindered due to the confinement effect of the concrete cover. Whereas, for GFRP bars exposed to simulated concrete pore solution, the erosion of GFRP bars by the corrosive solution occurs over the entire surface and  $H_2O/OH^-$  can freely move around the FRP bars without any obstacle. Furthermore, when the outer layer of the FRP bars loosen with time, the  $H_2O/OH^-$  ions can easily penetrate into the interior layer due to the lack of any confinement. In addition, the reductions in strength of GFRP bars reported by Sen et al. [24] are rather huge compared with those by other research results. This discrepancy is readily explained. First, GFRP bars used by Sen et al. were completely exposed to the simulated alkaline solution, while the specimens in other two studies were embedded in concrete. Further, the pH of alkaline solution used by Sen et al.



(pH = 13.5) was considerably higher than that of HP-SWSSC solution used in the present study (pH = 12.7).

### 3.2. Stress-strain curves and Young's modulus

The typical experimental stress-strain curves for BFRP and GFRP tested in different conditions are plotted in Fig. 6(a) and (b) respectively. It is clear that almost all the tensile stress-strain curves present a linear elastic behaviour up to failure, and the initial linear stage of each curve is close to each other. It is interesting to find that the curves of some specimens (e.g., B25/21/40% and GR) possess more than one peak. This may be due to the intrinsic inhomogeneity and uneven surface of FRP bars. As is known, the fibres inside FRP cannot fail simultaneously due to the Weibull statistics of strengths of fibres. The uneven surface of FRP bars also causes the stress concentration during the tensile test. Therefore, the failures of both BFRP and GFRP bars were accompanied by the separation/delamination of fibres and rupture of spiral bundles on the deformed surface of bars [15,25,27]. The separation/delamination of a large number of fibres at a time could cause the first peak. However, the remaining intact core of FRP bars continues to carry the increased load until the final rupture of FRP bars. Furthermore, Fig. 7 summarizes the tensile elastic modulus retention of BFRP and GFRP bars in all exposure conditions. The results show that the Young's modulus of both BFRP and GFRP bars under sustained loads are not affected by the HP-SWSSC exposure conditions. This is basically consistent with the modulus values of BFRP and GFRP bars in zero-stress condition exposed to SWSSC solutions [11]. Similar results were reported by Wu et al. [17] for BFRP bars in alkaline solution.

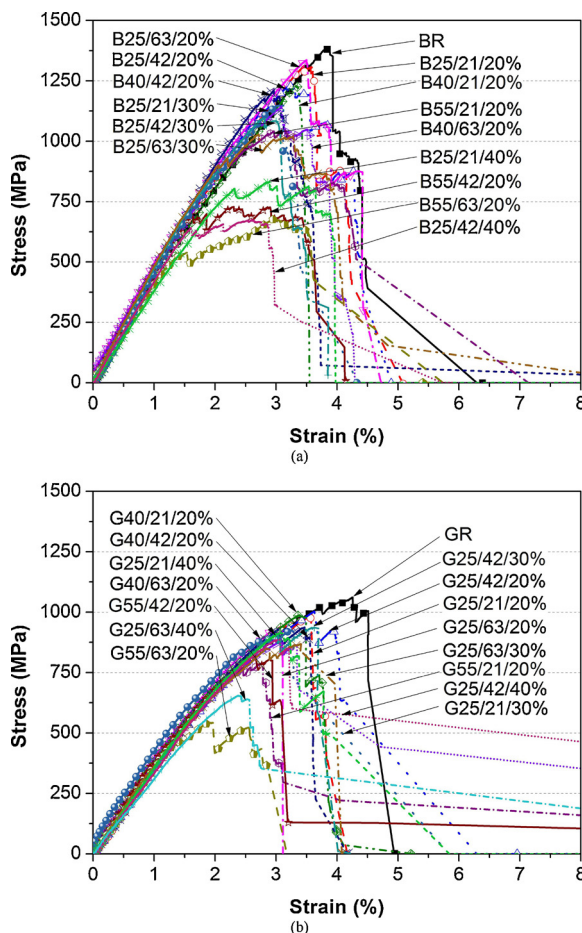


Fig. 6. Effect of conditioning in HP-SWSSC solution on tensile stress-strain curves of FRP bars: (a) BFRP and (b) GFRP.

## 4. Failure morphologies

### 4.1. Surface appearance

Changes in the surface appearance of BFRP and GFRP bars upon exposure to HP-SWSSC solution under different conditions are shown in Fig. 8. As shown in Fig. 8(a), for BFRP specimens at 20% stress level at 25 °C, there was no visible change with increasing exposure time from 21 to 63 days; whereas the changes were quite distinct when exposure temperature was increased from 25 °C to 40 °C, and then to 55 °C. The change was most pronounced at 55 °C when BFRP changed from black to grey, which is probably due to the reaction of basalt fibres with chloride in HP-SWSSC solution. In addition, at 25 °C, the surface of BFRP bars changed slightly at the relatively low stress levels (i.e., 20% and 30%). Though the colour change was not prominent, obvious macroscopic cracks occurred on the surface of BFRP bars at ~40% stress level. This is consistent with the dramatic drop in tensile strength retention of BFRP bars in the case of the 40% stress level (Fig. 3). As shown in Fig. 8(b), compared with BFRP bars, the surface colour change of GFRP bars under the same conditions was not obvious. This further confirms that the degradation of GFRP is much slower than that of BFRP in HP-SWSSC solution, which is consistent with the tensile test results (Figs. 3 and 4). This is also consistent with the previous results for BFRP and GFRP bars in zero-stress condition exposure to SWSSC solutions [11].

### 4.2. SEM analysis and degradation mechanism

The synergistic effect of combined stresses and traditional concrete environment on GFRP bars has been investigated by many previous researchers [23,35,36], and the detailed degradation mechanisms was

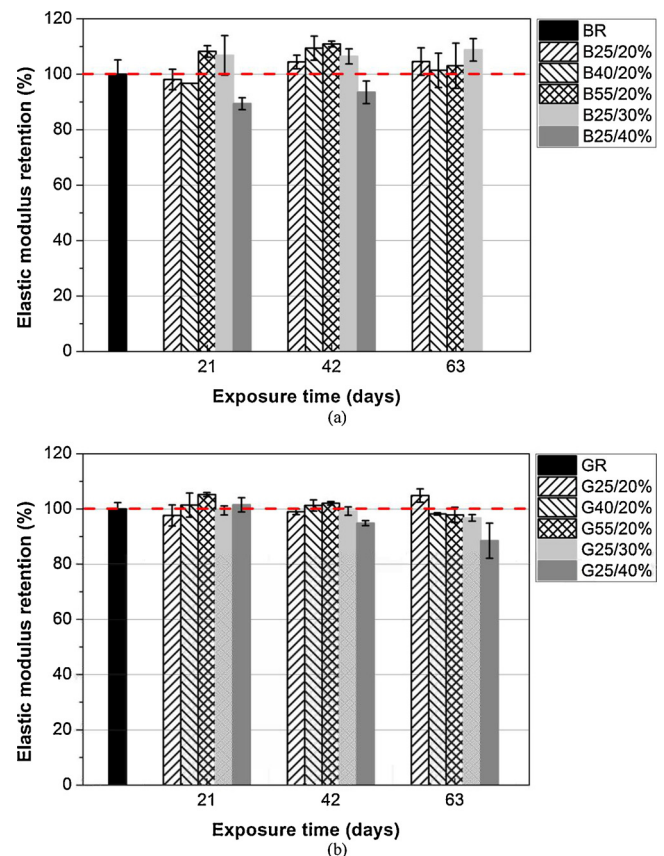


Fig. 7. Elastic modulus retention of conditioned FRP bars in HP-SWSSC solution: (a) BFRP and (b) GFRP.



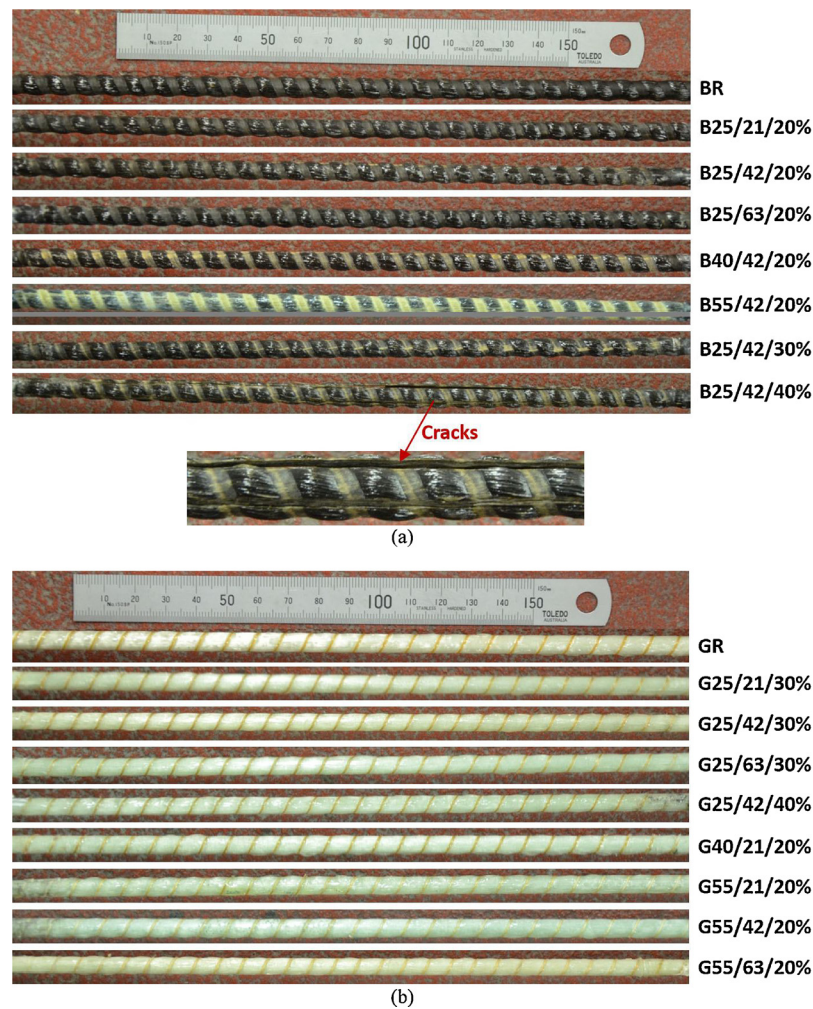


Fig. 8. Photographs of FRP bars after conditioned exposure to HP-SWSSC solution: (a) BFRP and (b) GFRP.

summarized in Ref. [41]. Our previous study showed that the degradation mechanisms of GFRP bars in SWSSC and traditional concrete environments were similar, while those of BFRP bars in two types of environments were much different due to the chemical reaction between  $\text{Cl}^-$  and  $\text{Fe}^{2+}$  [11]. In addition, considering that the BFRP bars are relatively less explored, this study laces more focus on the investigation of the degradation mechanism of BFRP bars under sustained load in HP-SWSSC environment. As schematically shown in Fig. 9, the critical areas of interest were selected in the cross-section for SEM images to illustrate the influence of three environmental factors (i.e., exposure time, exposure temperature and stress level) on the damage evolution along the cross section of BFRP bars. The detailed SEM images of the reference and conditioned BFRP specimens are presented in Fig. 10.

As shown in Fig. 10(a), in the case of an unexposed BFRP bar specimen (i.e., reference), the basalt fibre, resin and fibre/resin interface area are all intact without any sign of corrosion. Fig. 10(b)–(d) show the increasing extent of degradation with increasing exposure time of the BFRP bars at 20% stress. It is clear that for BFRP under the 20% stress level at 25 °C, the damaged parts are only limited to the outmost edge that was first to get in contact of the corrosive HP-SWSSC solution, and the main damage mode is the fibre/resin interface debonding with a small degree of fibre degradation. By comparing Fig. 10(e) and (c), it is obvious that the damaged area of BFRP at 20% stress in 42 days increased significantly when the temperature was increased from 25 °C to 40 °C. On a further increase in temperature (to 55 °C), some obvious interfacial cracks were generated that gradually developed towards the

specimen centre. For the effects of stress on the degradation of BFRP bars, Fig. 10(g) shows an interfacial cracks occurs in B25/42/30% specimen, and this proves that 30% stress level can cause the extension of interface debonding observed at 25 °C in HP-SWSSC solution. Fig. 10(h) clearly shows three huge macroscopic cracks in the cross section of B25/42/40% specimen. This proves that further increase in stress (to 40%) converts initial interface debonding to the catastrophic macroscopic cracks for BFRP bars in HP-SWSSC solution. Meanwhile, the damage in the outermost edge is still in the mode of interface debonding with bits of fibre corrosion. These SEM images prove that the combined effect of exposure temperature and stress can accelerate the generation and development of interface debonding/cracks in BFRP bars when exposed to HP-SWSSC solution. These results also reveal that the stress level is more effective than exposure temperature in accelerating the degradation of BFRP in SWSSC solution.

Furthermore, Fig. 11 compares the effects of different stress levels on degradation along the longitudinal section of BFRP bars after exposure to HP-SWSSC solution at 25 °C. Areas for SEM imaging were selected from the core of BFRP specimens, as identified by the boxes and arrows. As shown in Fig. 11(a), it is clear that the fibre, resin and interface area are all intact in the reference BFRP specimen. The SEM images for BFRP specimens exposed to HP-SWSSC solution at 20% (Fig. 11(b)) and 30% (Fig. 11(c)) stress levels appear similar to those of reference. However, as the stress level increased to 40%, some micro cracks in the fibre and resin areas occurred, as marked by the red enclosures in Fig. 11(d). These results again confirm that the 40% stress level is too high for BFRP bars exposed to HP-SWSSC solution.

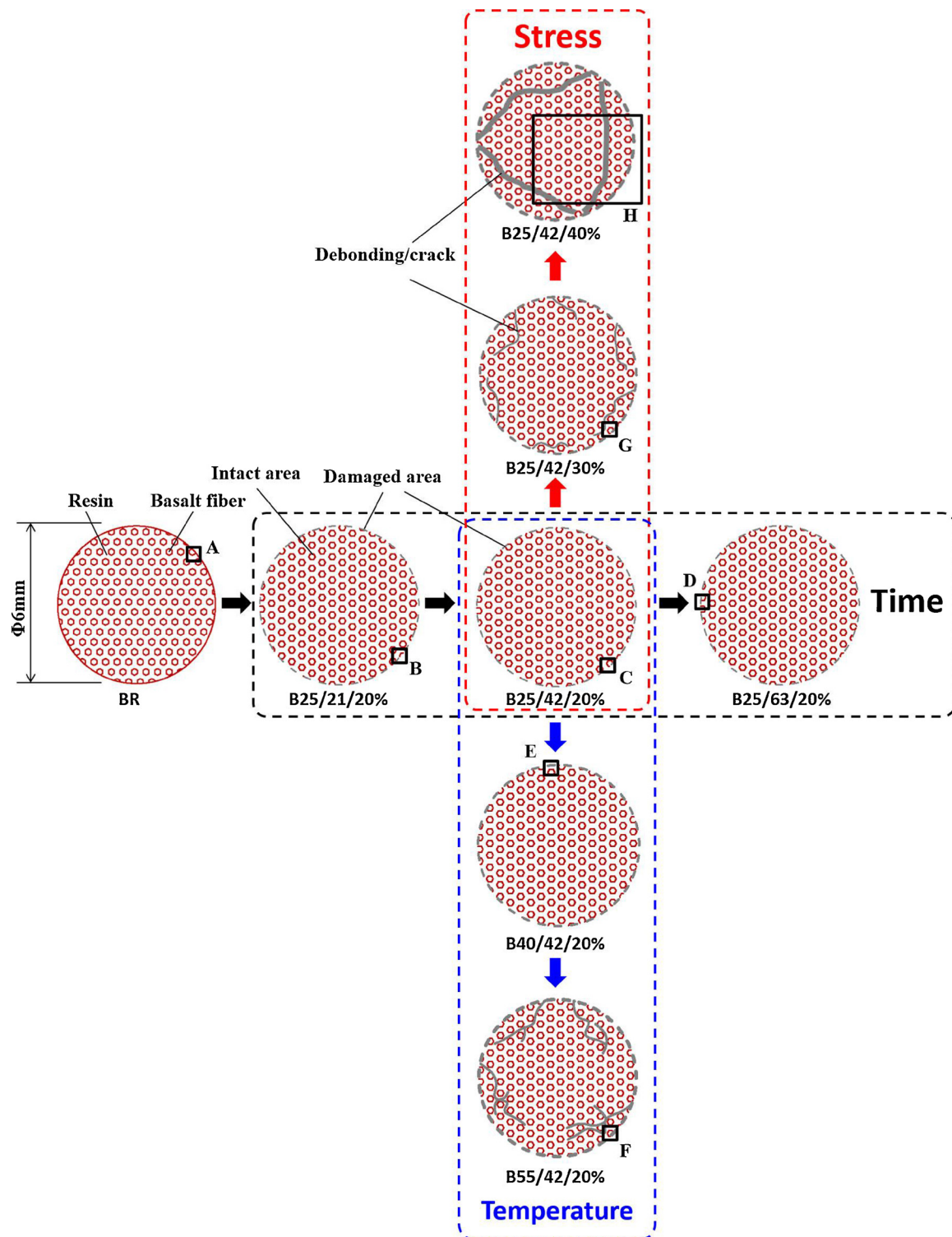


Fig. 9. Schematics of evolution of the damage along cross section of the BFRP bars after conditioned exposure to HP-SWSSC solution.

Finally it is interesting to compare the damage of BFRP specimens under zero-stress and stressed condition in SWSSC solution. As shown in Ref. [11], for zero-stress case, the corrosion front uniformly progressed from edge towards the centre of specimen, whereas for stressed case, as shown in Fig. 10(f), this front progressed irregularly due to debonding/cracks caused by the combined effect of exposure temperature and stress that dominated the degradation mode of the specimen. Such degradation would dramatically accelerate the penetration of SWSSC solution. As a result, the degradation of stressed FRP bars was much

quicker than that of the unstressed ones in the same corrosive conditions.

## 5. Prediction of long-term performance of bfrp bars

### 5.1. Existing models

To predict the long-term performance of FRP bars in civil engineering environment, the following three models based on the



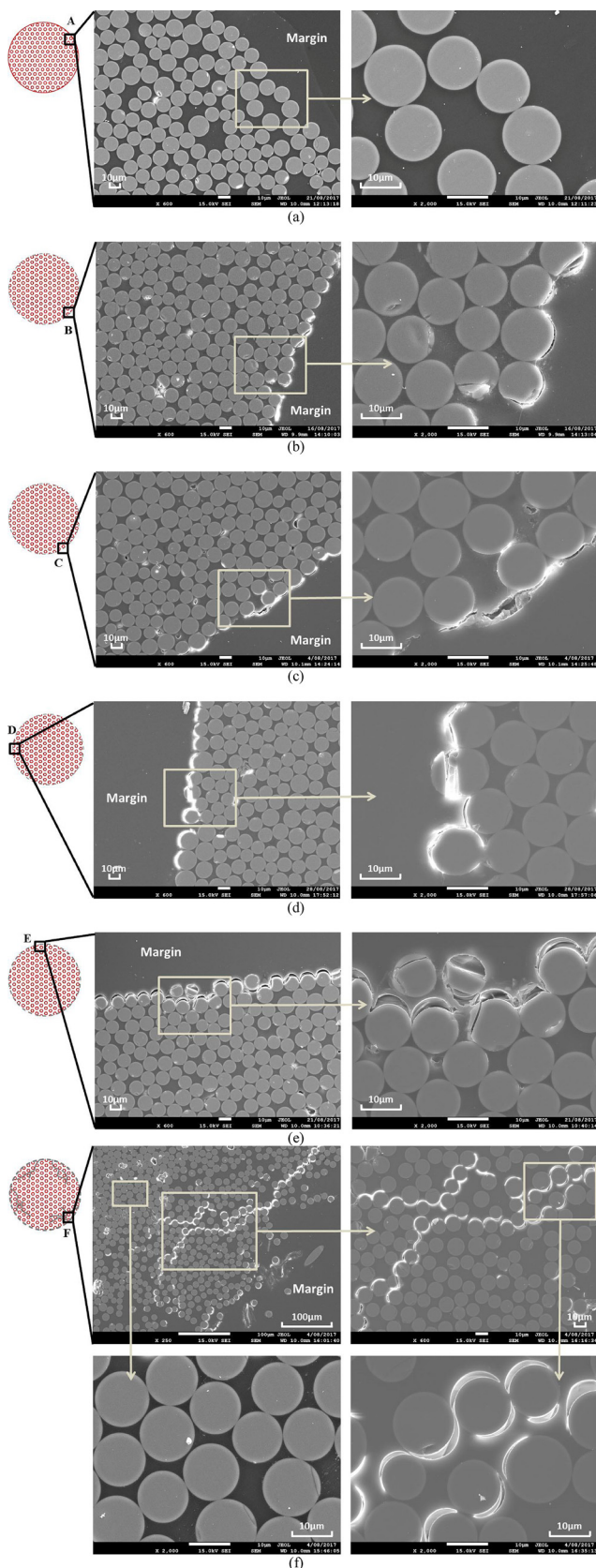


Fig. 10. SEM images for BFRP specimens after conditioned exposure to HP-SWSSC solution: (a) BR; (b) B25/21/20%; (c) B25/42/20%; (d) B25/63/20%; (e) B40/42/20%; (f) B55/42/20%; (g) B25/42/30% and (h) B25/42/40%.

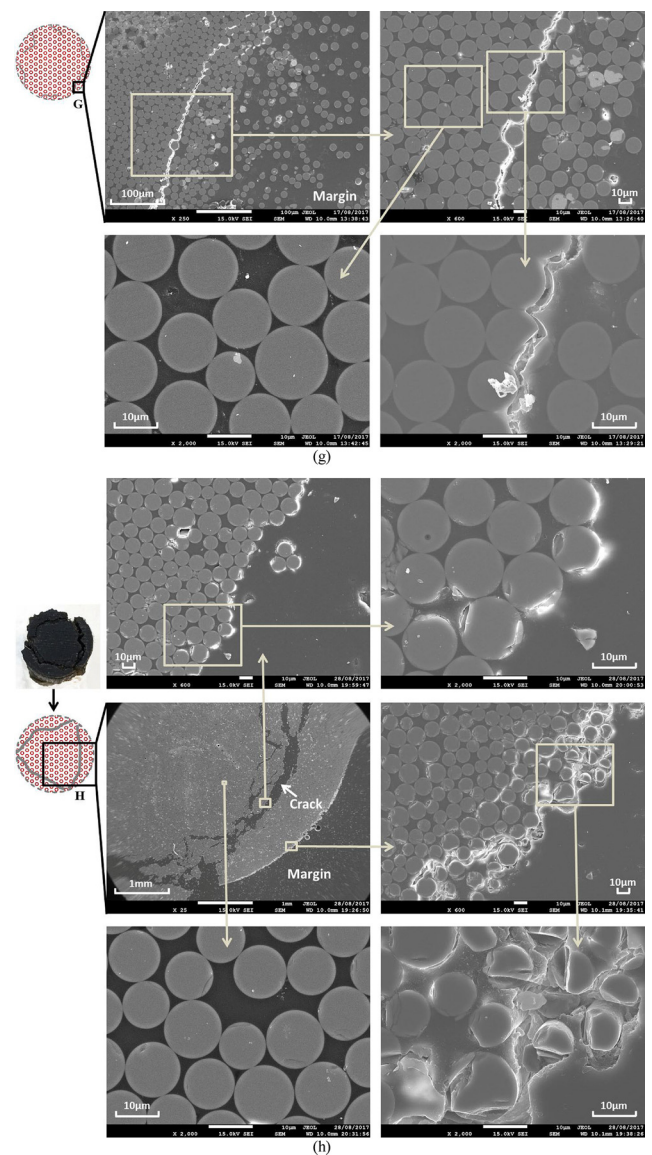


Fig. 10. (continued)

Arrhenius theory [55] were generally adopted by previous researchers to analyse the experimental accelerated short-term data of FRP specimens [11,17,27,29,31,33,56]:

Model 1 [29,33,56]:

$$Y = a \log(t) + b \quad (1)$$

Model 2 [11,17,27]:

$$Y = 100 \exp(-t/\tau) \quad (2)$$

Model 3 [11,31,57]:

$$Y = (100 - Y_{\infty}) \exp(-t/\tau) + Y_{\infty} \quad (3)$$

where  $Y$  is the tensile strength retention (%);  $t$  is the exposure time;  $a$  and  $b$  are the regression constants;  $\tau$  is the fitted parameter; and  $Y_{\infty}$  is the tensile strength retention (%) at exposure time of infinity.

There are some intrinsic limitations in Model 1 as described in [11,17], e.g., Model 1 is only a phenomenological representation of test data and never provides any hypothesis for the degradation mechanism. The degradation mechanism for Model 2 and Model 3 is assumed to be debonding at the fibre/matrix interface. The two models have been successfully used to predict the long-term tensile strength retention of BFRP bars under zero-stress condition while exposed to two kinds of

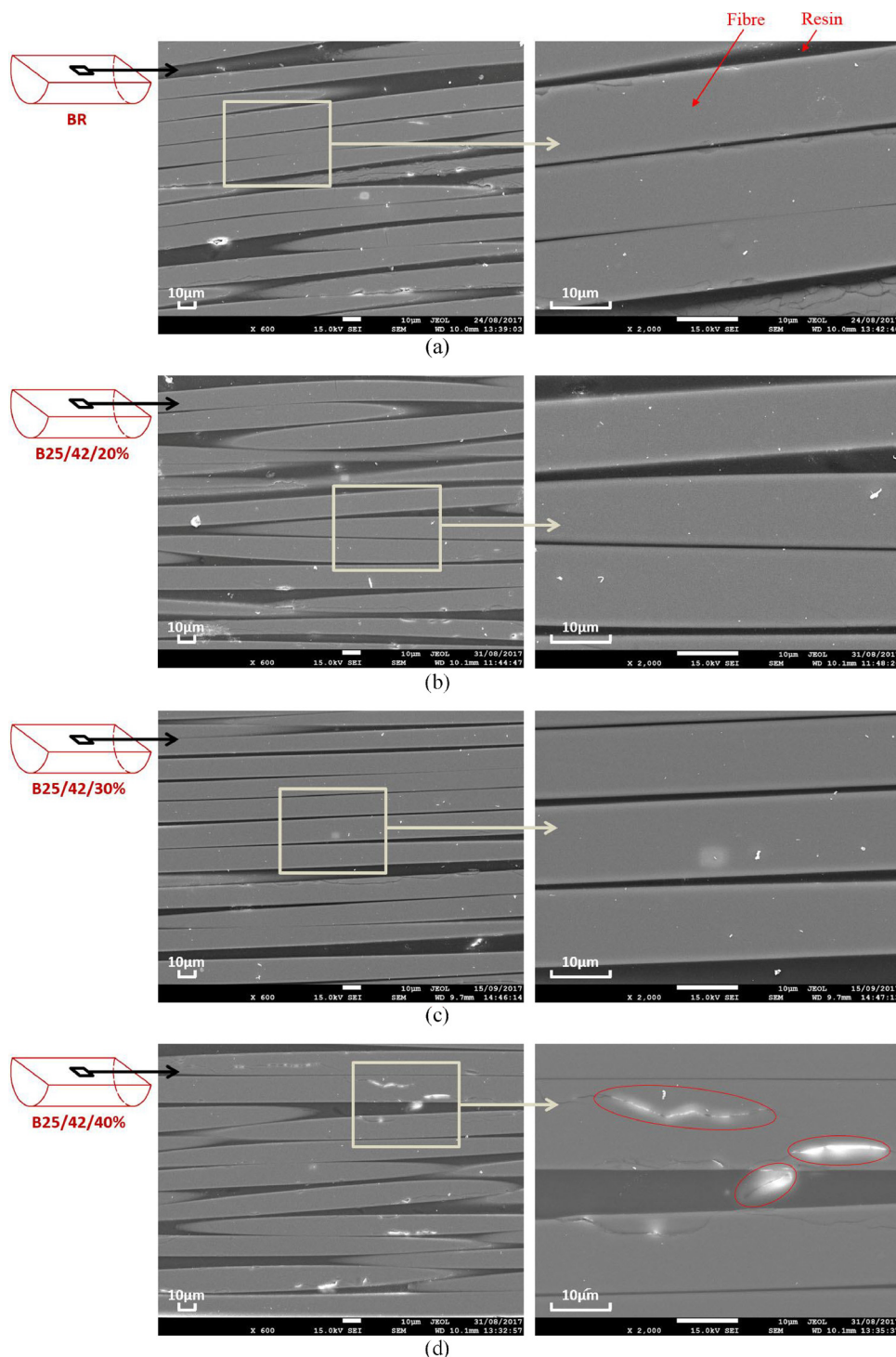


Fig. 11. SEM images along longitudinal section of the BFRP specimens after conditioned exposure to HP-SWSSC solution: (a) BR; (b) B25/42/20%; (c) B25/42/30% and (d) B25/42/40%.

SWSSC solutions environment [11]. Therefore, in this study, Model 2 and Model 3 will be used to evaluate the long-term tensile strength retention of BFRP bars under the condition of 20% stress level and exposure to HP-SWSSC solution. The effect of sustained load on the long-term durability of BFRP bars in SWSSC environment will also be discussed. As in our previous research [11], the locations of five serviced structures in Canada [28] were chosen as the case for prediction places in this study.

## 5.2. Arrhenius relation

The durability prediction procedure for Model 2 and Model 3 based on the Arrhenius relation is shown in Eq. (4) [10,11,17,31]:

$$k = A \exp(-E_a/RT) \quad (4)$$

where  $k$  is the degradation rate (1/time);  $A$  is a constant related to material and degradation process;  $E_a$  is the activation energy;  $R$  is the universal gas constant; and  $T$  is the temperature in Kelvin. The basic assumption of the Arrhenius relationship is that the single dominant



**Table 7**

Experimental tensile strength retention of BFRP bars exposed to HP-SWSSC solution used to develop the long-term performance models.

| Specimen type        | Exposure temperature (°C) | Tensile strength retention (%) |                   |                   |
|----------------------|---------------------------|--------------------------------|-------------------|-------------------|
|                      |                           | Aging for 21 days              | Aging for 42 days | Aging for 63 days |
| Unstressed BFRP [11] | 32                        | 99.3                           | 93.8              | 97.9              |
|                      | 40                        | 95.3                           | 93.0              | 90.2              |
|                      | 55                        | 89.7                           | 81.1              | 77.4              |
| Stressed BFRP        | 25                        | 92.4                           | 94.0              | 92.9              |
|                      | 40                        | 88.4                           | 89.8              | 81.7              |
|                      | 55                        | 69.6                           | 59.8              | 55.1              |

degradation mechanism of the material does not change with time and temperature during the exposure, whereas the rate of degradation is accelerated with the increase in temperature. Furthermore, Eq. (4) can be transformed into:

$$\frac{1}{k} = \frac{1}{A} \exp(E_a/RT) \quad (5)$$

$$\ln\left(\frac{1}{k}\right) = \frac{E_a}{R} \frac{1}{T} - \ln A \quad (6)$$

From Eq. (5), the degradation rate  $k$  can be expressed as the inverse of the time needed for a material property to reach a given value [29]. From Eq. (6), one can further observe that the logarithm of time needed for a material property to reach a given value is a linear function of  $1/T$  with the slope of  $E_a/R$  value [29].

According to Eq. (4), for two different exposure temperatures, the time-shift factor (TSF) for a range of high temperature ( $T_1$ ) to low temperature ( $T_0$ ) can be calculated as:

$$\text{TSF} = \frac{t_0}{t_1} = \frac{c/k_0}{c/k_1} = \frac{k_1}{k_0} = \frac{A \exp(-E_a/RT_1)}{A \exp(-E_a/RT_0)} = \exp\left[\frac{E_a}{R} \left(\frac{1}{T_0} - \frac{1}{T_1}\right)\right] \quad (7)$$

where  $t_0$  and  $t_1$  are the time required for the tensile strength to reach a given retention value (represented by the constant  $c$ ) at temperatures of  $T_0$  and  $T_1$ , respectively;  $k_0$  and  $k_1$  are the degradation rates at temperatures of  $T_0$  and  $T_1$  respectively.

Based on the discussion above, the long-term tensile performance of BFRP bars in HP-SWSSC solution can be predicted at any given temperature.

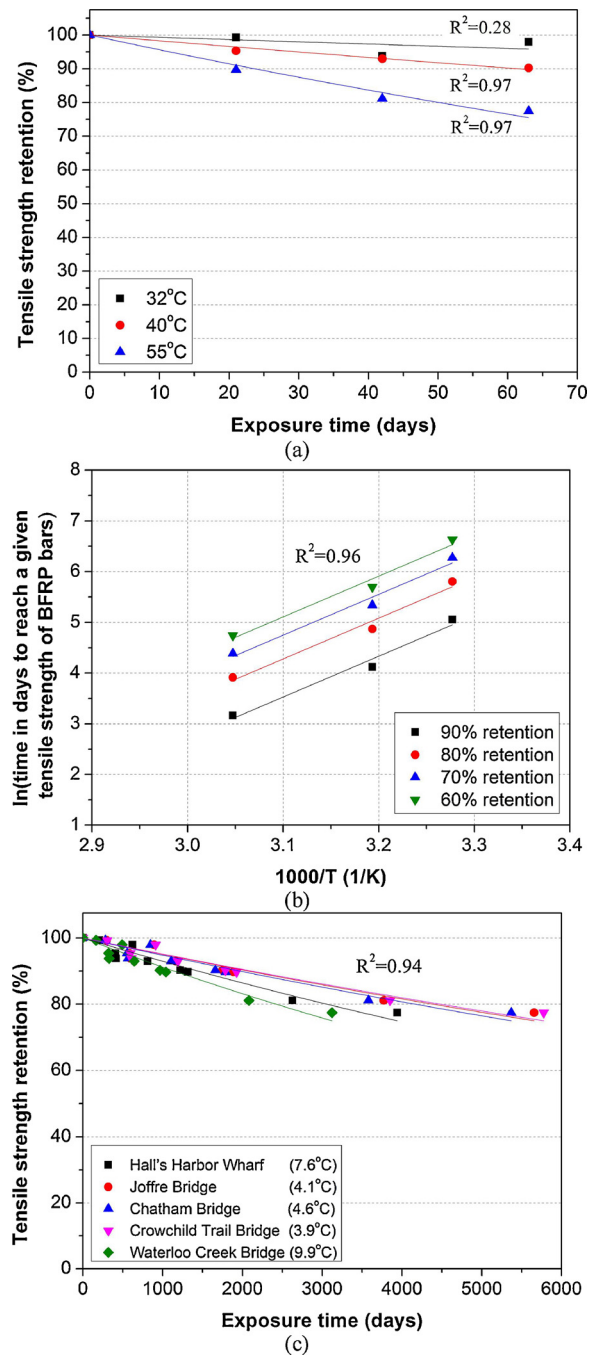
### 5.3. Prediction procedure

All the tensile strength retention data of BFRP bars exposed to HP-SWSSC solution that were used for developing the long-term performance models are listed in Table 7. The prediction procedures of unstressed BFRP and stressed BFRP bars in HP-SWSSC solution using Model 2 and Mode 3 are similar. In case of unstressed BFRP specimens using Model 2, the prediction procedure is as follows:

Firstly, Eq. (2) was fitted to the test results given in Fig. 12(a) to obtain  $\tau$  values and correlation coefficients ( $R^2$ ) for different exposure temperatures as shown in Table 8.

Secondly, the Arrhenius plots, as shown in Fig. 12(b), were drawn. The time ( $t$ ) needed for tensile strength retention to reach 60, 70, 80, and 90% at different exposure temperatures (i.e., 32, 40 and 55 °C) were obtained from Eq. (2) with regression coefficient  $\tau$  listed in Table 8. Then, Eq. (6) was fitted to those values ( $\ln(1/k) = \ln t$  and  $1/T$ ) as shown in Fig. 12(b), and the straight lines were parallel to each other. The slopes of the straight lines ( $E_a/R$ ), the calculated parameters ( $E_a$  and  $A$ ), and the correlation coefficients ( $R^2$ ) are shown in Table 9.

Thirdly, from Eq. (7), the TSF with reference temperatures  $T_0$  equal to annual mean temperature of each of the chosen Canada structures



**Fig. 12.** Prediction of unstressed BFRP bars in HP-SWSSC solution using Model 2: (a) fitted curves; (b) Arrhenius plots of tensile strength retention and (c) master curves.

are obtained and listed in Table 10.

Finally, the exposure times at different temperatures (i.e., 32, 40 and 55 °C) shown in Fig. 12(a) were multiplied by corresponding TSF values, and transformed results are shown in Fig. 12(c). Next, by fitting Eq. (2) to data in Fig. 12(c), the master curves shown were defined for unstressed BFRP bars in HP-SWSSC solution at different prediction temperatures. The fitted  $\tau$  values and correlation coefficients ( $R^2$ ) are shown in Table 11.

The prediction of unstressed BFRP specimens using Model 3 fully follows the above steps as for Model 2, only except for changing Eqs. (2) and (3).

The prediction procedures of stressed BFRP specimens using Model 2 and Model 3 are same as those of unstressed BFRP specimens as

**Table 8**  
Coefficients of regression in Eqs. (2) and (3) for BFRP bars in HP-SWSSC solution.

| Specimen type   | Temperature (°C) | Model 2 (fitted parameters in Eq. (2)) |       | Model 3 (fitted parameters in Eq. (3)) |              |       |
|-----------------|------------------|----------------------------------------|-------|----------------------------------------|--------------|-------|
|                 |                  | $\tau$                                 | $R^2$ | $\tau$                                 | $Y_{\infty}$ | $R^2$ |
| Unstressed BFRP | 32               | 1487.1                                 | 0.28  | 458.9                                  | 67.6         | 0.29  |
|                 | 40               | 584.2                                  | 0.97  | 169.3                                  | 67.6         | 0.98  |
|                 | 55               | 224.7                                  | 0.97  | 51.5                                   | 67.6         | 0.99  |
| Stressed BFRP   | 25               | 686.4                                  | 0.36  | 302.0                                  | 53.4         | 0.39  |
|                 | 40               | 308.9                                  | 0.81  | 127.6                                  | 53.4         | 0.83  |
|                 | 55               | 87.3                                   | 0.90  | 20.2                                   | 53.4         | 0.99  |

mentioned above. In addition, it is noted that for Model 3, when the data of Fig. 13(a) or Fig. 15(a) were fitted by Eq. (3), the  $Y_{\infty}$  was first confirmed according to fitting the data at 55 °C case, and then the rest of the data (i.e., 25 °C, 32 °C, 40 °C) were fitted with the obtained  $Y_{\infty}$ .

All the detailed prediction parameters for unstressed BFRP and stressed BFRP (at 20% stress level) in HP-SWSSC solution using two models are summarized in Figs. 12–15 and Tables 8–11.

#### 5.4. Prediction results

The prediction results of the unstressed and stressed BFRP bars (at 20% stress level) in HP-SWSSC solution at 9.9 °C using Model 2 are presented in Fig. 16. Fig. 16(a) and Table 9 show that  $E_a/R$  of the stressed BFRP bars is 6696, which is a little lower than 8039 for the unstressed BFRP bars. The similar  $E_a/R$  values for BFRP and GFRP bars in different corrosive environments were also obtained using Model 2.

**Table 9**  
Coefficients of regression in Eq. (6) for Arrhenius plots in Model 2 and Model 3.

| Specimen type   | Tensile strength retention (%) | Model 2 |                |             |       | Model 3 |                |             |       |
|-----------------|--------------------------------|---------|----------------|-------------|-------|---------|----------------|-------------|-------|
|                 |                                | $E_a/R$ | $E_a$ (kJ/mol) | $A$ (1/day) | $R^2$ | $E_a/R$ | $E_a$ (kJ/mol) | $A$ (1/day) | $R^2$ |
| Unstressed BFRP | 90                             | 8039    | 66.8           | $e^{21.4}$  | 0.96  | 9369    | 77.9           | $e^{25.65}$ | 0.98  |
|                 | 80                             | 8039    | 66.8           | $e^{20.64}$ | 0.96  | 9369    | 77.9           | $e^{24.70}$ | 0.98  |
|                 | 70                             | 8039    | 66.8           | $e^{20.17}$ | 0.96  | 9369    | 77.9           | $e^{23.70}$ | 0.98  |
|                 | 60                             | 8039    | 66.8           | $e^{19.81}$ | 0.96  | –       | –              | –           | –     |
| Stressed BFRP   | 90                             | 6696    | 55.6           | $e^{18.09}$ | 0.95  | 8763    | 72.9           | $e^{24.93}$ | 0.89  |
|                 | 80                             | 6696    | 55.6           | $e^{17.34}$ | 0.95  | 8763    | 72.9           | $e^{24.08}$ | 0.89  |
|                 | 70                             | 6696    | 55.6           | $e^{16.87}$ | 0.95  | 8763    | 72.9           | $e^{23.47}$ | 0.89  |
|                 | 60                             | 6696    | 55.6           | $e^{16.51}$ | 0.95  | 8763    | 72.9           | $e^{22.83}$ | 0.89  |

Note: “–” represents “not available”.

**Table 10**  
Time-shift factor for BFRP bars in HP-SWSSC solution.

| Specimen type   | Model   | Temperature (°C) | Time-shift factor (TSF)       |                         |                          |                                  |                                 |
|-----------------|---------|------------------|-------------------------------|-------------------------|--------------------------|----------------------------------|---------------------------------|
|                 |         |                  | Hall's Harbor Wharf<br>7.6 °C | Joffre Bridge<br>4.1 °C | Chatham Bridge<br>4.6 °C | Crowchild Trail Bridge<br>3.9 °C | Waterloo Creek Bridge<br>9.9 °C |
| Unstressed BFRP | Model 2 | 32               | 9.9                           | 14.2                    | 13.4                     | 14.5                             | 7.8                             |
|                 |         | 40               | 19.3                          | 27.8                    | 26.4                     | 28.4                             | 15.3                            |
|                 |         | 55               | 62.6                          | 89.8                    | 85.2                     | 91.7                             | 49.6                            |
|                 | Model 3 | 32               | 14.4                          | 22.0                    | 20.7                     | 22.5                             | 11.0                            |
|                 |         | 40               | 31.6                          | 48.1                    | 45.3                     | 49.3                             | 24.1                            |
|                 |         | 55               | 124.0                         | 189.0                   | 177.8                    | 193.6                            | 94.6                            |
| Stressed BFRP   | Model 2 | 25               | 4.0                           | 5.4                     | 5.2                      | 5.5                              | 3.3                             |
|                 |         | 40               | 11.8                          | 15.9                    | 15.3                     | 16.2                             | 9.7                             |
|                 |         | 55               | 31.3                          | 42.4                    | 40.6                     | 43.1                             | 25.8                            |
|                 | Model 3 | 25               | 6.2                           | 9.2                     | 8.7                      | 9.4                              | 4.8                             |
|                 |         | 40               | 25.3                          | 37.5                    | 35.4                     | 38.3                             | 19.6                            |
|                 |         | 55               | 90.8                          | 134.6                   | 127.2                    | 137.7                            | 70.5                            |

**Table 11**  
Coefficients of the regression equations for the master curves of BFRP bars in HP-SWSSC solution.

| Specimen type   | Structure              | Mean annual temperature (°C) | Model 2 |       | Model 3 |              |       |
|-----------------|------------------------|------------------------------|---------|-------|---------|--------------|-------|
|                 |                        |                              | $\tau$  | $R^2$ | $\tau$  | $Y_{\infty}$ | $R^2$ |
| Unstressed BFRP | Hall's Harbor Wharf    | 7.6                          | 13698.4 | 0.94  | 6148.5  | 67.6         | 0.96  |
|                 | Joffre Bridge          | 4.1                          | 19663.3 | 0.94  | 9369.8  | 67.6         | 0.96  |
|                 | Chatham Bridge         | 4.6                          | 18663.3 | 0.94  | 8816.9  | 67.6         | 0.96  |
|                 | Crowchild Trail Bridge | 3.9                          | 20079.2 | 0.94  | 9601.0  | 67.6         | 0.96  |
|                 | Waterloo Creek Bridge  | 9.9                          | 10854.6 | 0.94  | 4688.3  | 67.6         | 0.96  |
| Stressed BFRP   | Hall's Harbor Wharf    | 7.6                          | 2898.6  | 0.90  | 2326.4  | 53.4         | 0.94  |
|                 | Joffre Bridge          | 4.1                          | 3916.8  | 0.90  | 3449.9  | 53.4         | 0.94  |
|                 | Chatham Bridge         | 4.6                          | 3750.3  | 0.90  | 3259.1  | 53.4         | 0.94  |
|                 | Crowchild Trail Bridge | 3.9                          | 3985.8  | 0.90  | 3529.5  | 53.4         | 0.94  |
|                 | Waterloo Creek Bridge  | 9.9                          | 2387.9  | 0.90  | 1805.2  | 53.4         | 0.94  |

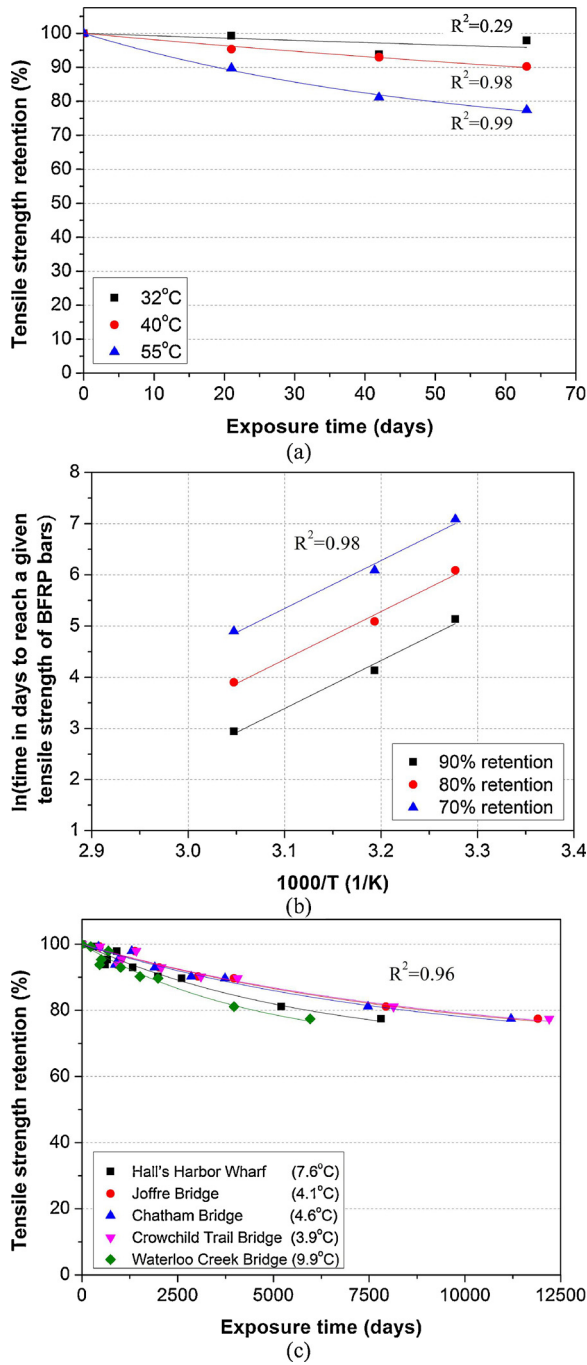


Fig. 13. Prediction of unstressed BFRP bars in HP-SWSSC solution using Model 3: (a) fitted curves; (b) Arrhenius plots of tensile strength retention and (c) master curves.

Wu et al. [17] reported the  $E_a/R$  value of 7519 for the same BFRP bars exposed to alkaline solution. Chen et al. [27] obtained two  $E_a/R$  values of 1420 and 4890 for two types of GFRP bars (with different types of E-glass fibres) exposed to simulated concrete pore solution, respectively. Deike [58] obtained an  $E_a/R$  value of 6300 for GFRP bars embedded in moist concrete. Since the different  $E_a/R$  resulted in different degradation rates and probably different mechanisms of FRP bars [27], this study fills the research gap of FRP bars in SWSSC environment. As shown in Fig. 16(b), the degradation rate of stressed BFRP bars was clearly higher than that of unstressed BFRP bars. This prediction results are consistent with the short-term experimental observation.

The prediction results of unstressed and stressed BFRP bars in HP-

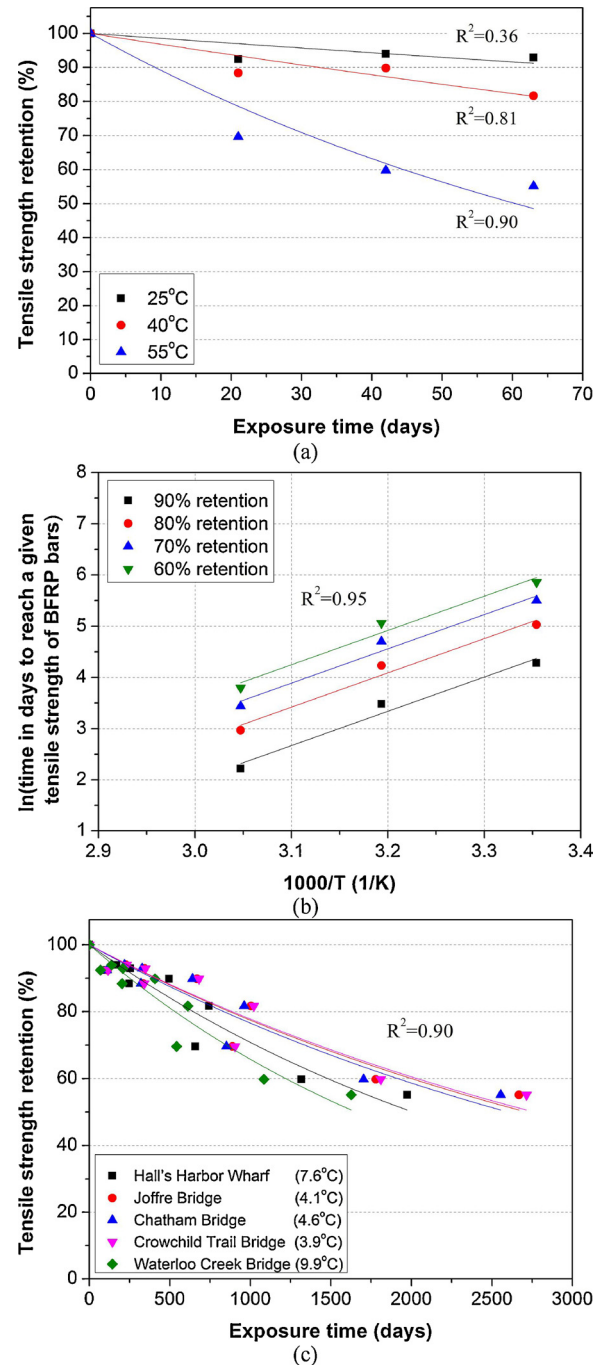


Fig. 14. Prediction of stressed BFRP bars under 20% stress level in HP-SWSSC solution using Model 2: (a) fitted curves; (b) Arrhenius plots of tensile strength retention and (c) master curves.

SWSSC solution at 9.9°C using Model 3 are presented in Fig. 17 and Table 11. The results of Model 3 show that the long-term tensile strength retention of unstressed BFRP converges to 67.6%, while that of stressed BFRP converges to 53.4%. Similarly, Davalos et al. [31] also reported that the long-term retentions ( $Y_\infty$ ) of GFRP bars in saturated concrete environment are 45% for not loaded and 38% for loaded (at ~20% stress level) cases respectively. These predictions show that 20% stress level effectively reduces the long-term life for FRP bars in various concrete environments.

In the guideline of ACI 440.1R-15 [45], the suggested environmental reduction factor  $C_E$  for GFRP bars in concrete exposed to earth and weather is 70%, while the  $C_E$  for BFRP bars is not given.



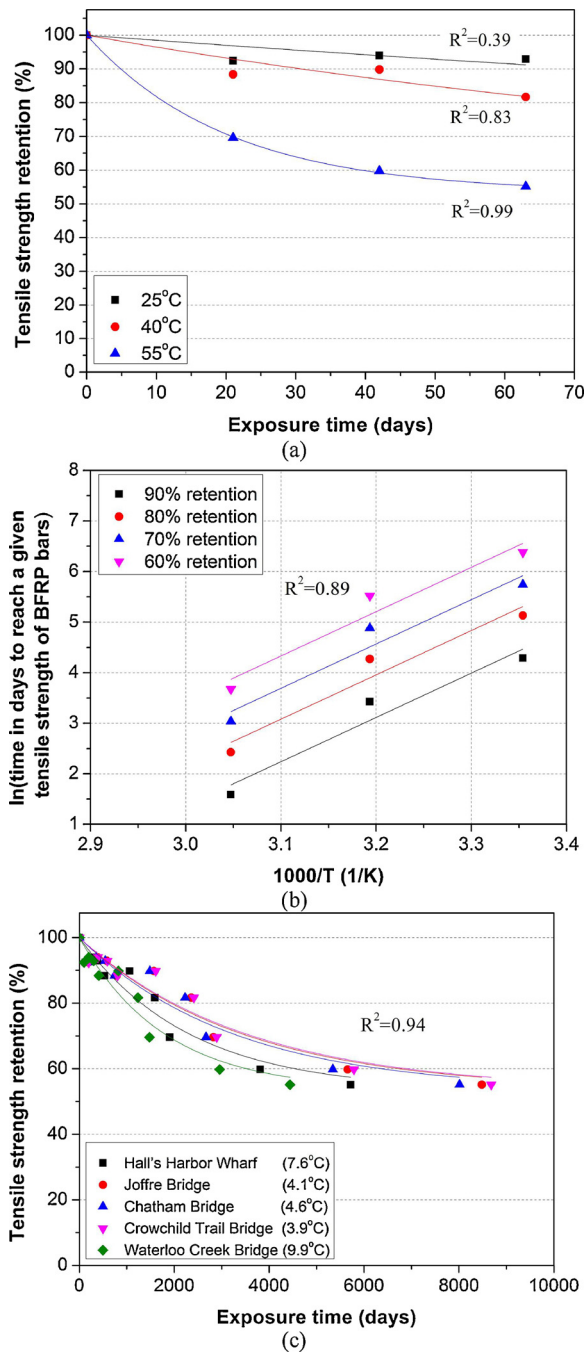


Fig. 15. Prediction of stressed BFRP bars under 20% stress level in HP-SWSSC solution using Model 3: (a) fitted curves; (b) Arrhenius plots of tensile strength retention and (c) master curves.

Accordingly, the predicted time to reach 70% retention for unstressed and stressed BFRP bars exposed to HP-SWSSC solution in five different Canadian locations is also calculated using Model 2 and Model 3 and summarized in Table 12. As shown in Table 12, in the case of Model 2, after exposure of 10.6–19.6 years for unstressed BFRP bars (while 2.3 to 3.9 years for stressed BFRP bars in HP-SWSSC solution), the tensile strength retention reaches the 70%. In the case of Model 3, after exposure of 33.4 to 68.5 years for unstressed BFRP bars (while 5.1 to 10.0 years for stressed BFRP bars), the tensile strength retention reaches the 70%. This shows the prediction results of Model 2 are much more conservative than those of Model 3.

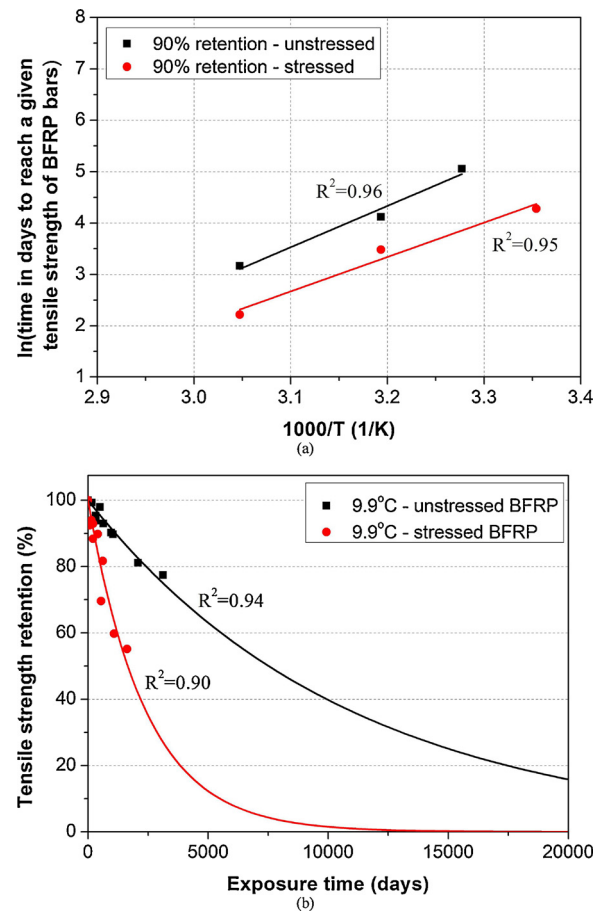


Fig. 16. Comparison of predictions for BFRP bars in HP-SWSSC solution at 9.9°C using Model 2: (a) Arrhenius plots of tensile strength retention and (b) master curves.

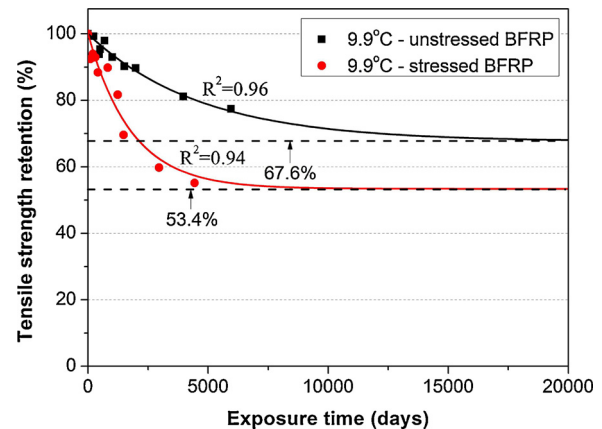


Fig. 17. Comparison of predictions for BFRP bars in HP-SWSSC solution at 9.9°C using Model 3.

### 5.5. Limitation and future research

Finally, for comparison with the field data of FRP bars, the tensile strength retentions of two kinds of BFRP bars after given serviced years from Model 2 and Model 3 were produced as shown in Table 13. According to the research of Mufti et al. [28], there were no changes in the microstructures of GFRP bars after 5–8 years of service in field concrete structures. Therefore, in this study, the tensile strength retentions of GFRP bars in five Canadian field structures were considered to be still at 100% level after 5–8 years. From the comparison of results in Table 13,

**Table 12**

Long-term predication results of BFRP bars at different mean annual temperatures using Model 2 and Model 3.

| Structure              | Mean annual temperature (°C) | Time in years to reach 70% tensile retention of BFRP bars |                                    |                                      |                                    |
|------------------------|------------------------------|-----------------------------------------------------------|------------------------------------|--------------------------------------|------------------------------------|
|                        |                              | Model 2                                                   |                                    | Model 3                              |                                    |
|                        |                              | Unstressed BFRP in HP-SWSSC solution                      | Stressed BFRP in HP-SWSSC solution | Unstressed BFRP in HP-SWSSC solution | Stressed BFRP in HP-SWSSC solution |
| Hall's Harbor Wharf    | 7.6 <sup>a</sup>             | 13.4                                                      | 2.8                                | 43.8                                 | 6.6                                |
| Joffe Bridge           | 4.1                          | 19.2                                                      | 3.8                                | 66.8                                 | 9.8                                |
| Chatham Bridge         | 4.6                          | 18.2                                                      | 3.7                                | 62.9                                 | 9.2                                |
| Crowchild Trail Bridge | 3.9                          | 19.6                                                      | 3.9                                | 68.5                                 | 10.0                               |
| Waterloo Creek Bridge  | 9.9                          | 10.6                                                      | 2.3                                | 33.4                                 | 5.1                                |

Note: a: annual mean temperature from the World Climate & Temperature online database by 12 August 2016. (<http://www.climateemps.com/>).

**Table 13**

Comparison between long-term prediction results and field results of FRP bars using Model 2 and Model 3.

| Structure              | Serviced years by 2007 | Mean annual temperature (°C) | Calculated tensile strength retention of FRP bars after the given serviced years (%) |                                    |                                      |                                    |                                                 |
|------------------------|------------------------|------------------------------|--------------------------------------------------------------------------------------|------------------------------------|--------------------------------------|------------------------------------|-------------------------------------------------|
|                        |                        |                              | Model 2                                                                              |                                    | Model 3                              |                                    | Retention of field GFRP (%) (Mufti et al. [28]) |
|                        |                        |                              | Unstressed BFRP in HP-SWSSC solution                                                 | Stressed BFRP in HP-SWSSC solution | Unstressed BFRP in HP-SWSSC solution | Stressed BFRP in HP-SWSSC solution |                                                 |
| Hall's Harbor Wharf    | 5                      | 7.6 <sup>a</sup>             | 87.5                                                                                 | 53.3                               | 91.6                                 | 74.7                               | 100 <sup>b</sup>                                |
| Joffre Bridge          | 7                      | 4.1                          | 87.8                                                                                 | 52.1                               | 92.2                                 | 75.6                               | 100                                             |
| Chatham Bridge         | 8                      | 4.6                          | 85.5                                                                                 | 45.9                               | 90.8                                 | 72.4                               | 100                                             |
| Crowchild Trail Bridge | 8                      | 3.9                          | 86.5                                                                                 | 48.1                               | 91.4                                 | 73.8                               | 100                                             |
| Waterloo Creek Bridge  | 6                      | 9.9                          | 81.7                                                                                 | 40.0                               | 87.8                                 | 67.3                               | 100                                             |

Note: a: annual mean temperature from the World Climate & Temperature online database by 12 August 2016. (<http://www.climateemps.com/>).

b: the retention 100% was inferred from the micro observation results.

it seems that the prediction results of Model 2 and Model 3 may be too conservative. The main reason for the limitation of the existing models is that the tested BFRP bars in this study were completely immersed in the simulated HP-SWSSC solution, which possess much higher humidity than the real serviced concrete structure [54,59]. In addition, the real temperature inside the concrete may be less than the atmospheric ambient temperature and may not be constant as assumed in the prediction model. A more comprehensive long-term prediction model for FRP bars in concrete structure is being investigated through considering the real humidity and temperature range inside concrete.

## 6. Conclusions

Based on the investigation on durability of BFRP and GFRP bars under different stress levels in simulated HP-SWSSC solution, the following conclusions can be drawn in this study:

- (1) The tensile strength retention of BFRP and GFRP bars is significantly affected by the combined effect of sustained stress and exposure temperature of test environment of HP-SWSSC solution, while the Young's modulus of the bars remained unchanged. Future research is required to investigate some key properties (such as  $T_g$ , fibre content) of FRP bars before and after conditioning.
- (2) Under the same exposure condition (i.e. stress level, exposure time and temperature), the tensile strength retention of BFRP bars is obviously lower than that of GFRP. This may be due to two reasons: (a) the higher moisture absorbability of BFRP than GFRP causes greater diffusion of  $H_2O$ ,  $OH^-$ ,  $Cl^-$  into BFRP bars, and (b) the chemical reaction between  $Cl^-$  and  $Fe^{2+}$  that is exclusive to basalt fibre (because of its Fe-oxide content) further accelerates the

degradation of BFRP bars in HP-SWSSC solution.

- (3) SEM results reveal the degradation mechanism of BFRP bars in stress condition in SWSSC solution, i.e., the development of debonding/cracks at the fibre/resin interface in the edge areas of stressed BFRP. Further investigation such as those using X-Ray CT scanning imaging may provide more insight into the degradation mechanism.
- (4) The prediction results show that the long-term tensile strength retention of stressed BFRP bars (20% stress level) was much lower than that of unstressed BFRP bars in HP-SWSSC environment. However, the current existing prediction models are still conservative. Further research is needed to improve the model by considering more realistic moisture content and internal temperature distribution of SWSSC, and degradation mechanism of FRP under the combined effect of corrosive environment and stresses.

## Acknowledgements

The authors wish to acknowledge the financial support provided by the Australian Research Council (ARC) through an ARC Discovery Grant (DP160100739), the Key Project of Chinese Ministry of Education (Grant No: 113029A) and the National Natural Science Foundation of China (Grant No: 51478145). The first author is grateful to the China Scholarship Council (CSC) for the support of his visiting to Monash University, Australia. The tests were conducted in the Civil Engineering Laboratory at Monash University with help provided by Mr. Long Goh, Mr. Saravanan Mani and Mr. Philip Warnes. We thank Mr. Dong Zhiqiang of Southeast University for his assistance to obtain BFRP bars.

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